PTC PHY002 期末资料

PTC Finals Prep Materials for PHY002

PTC 伴学社

2025.5.12

判天地之美，析万物之理

前言　preface

致每一位拿到这份资料的同学：

欢迎阅读 PTC PHY002 期末复习资料！

我们的资料根据所学章节专题展开，分为章节词汇、基本思想与公式、例题三部分，在此基础上加入了我们自己的辅助理解与解题思路，能够帮助同学在巩固基础的同时加强对知识的理解、习题的掌握。

这份期末资料经过了反复打磨，最后精简成了你手中的这份资料。这离不开我们物理组每一位成员的付出，使我们的资料得以成形；也得益于 PTC 每一位成员的帮助，使我们资料的精准度和完美度不断提高。

我们始终认为，学习物理不仅是记忆定律与演算习题，更是拥有一双看见本质的眼睛。从彩虹中看见光的色散，从潮汐中看见月的引力，从物理中看见万物的逻辑，折射着人类对真理的执着与浪漫。当你们为解题蹙眉时，请记住，每一次困惑与顿悟都是一场跨越时空的思维共振。愿你们在此间，找到属于自己的诗篇。

预祝各位阅读愉快，复习顺利，在期末考试中取得惊喜的成绩！

PTC 物理组全员——

陈朱恺　丁彦博　毛一茜　梅奕桐　邱语　滕启睿　张博涵乔

（按拼音首字母排序，顺序不分先后）

2025.5.12

简介、测量、估算

本节物理词汇：

Uncertainty 不确定度
Significant figures 有效数字
Order of magnitude 数量级
Dimensional analysis 量纲分析
SI Units 国际单位制单位
Scientific notation 科学记数法
Converting units 单位换算
Dimensions 量纲

【基本思想】

1.Measurement & Uncertainty

All measurements have uncertainty due to instrument limits. Use significant figures to indicate reliable digits.

2.Scientific Notation & SI Units

Express values clearly with scientific notation. Use SI units (meters, kilograms) and prefixes for consistency.

3.Order-of-Magnitude Estimates

Simplify calculations by rounding to 1 significant figure.

4.Dimensional Analysis

Verify equations by checking dimensions. Mismatched dimensions reveal errors.

【辅助理解】

1. 所有测量都有不确定度，需用有效数字记录可靠数值
   “23.21 cm”：四位有效数字

2. 用科学记数法（如 3.69×10^4）明确有效数字位数；

3. 国际单位制（SI）（如米、千克）及其前缀（如千“k”、毫“m”）是物理量的统一标准。

4. 数量级估算通过保留一位有效数字快速得到近似结果

   1 km 直径、10 m 深的湖，水量约为 $10^7\mathrm{m}^3$

5. 检查方程各物理量的量纲（如长度 L、时间 T）是否一致，避免错误（如速度公式中不能将“速度”与“长度”相加）。

一维空间中的运动学

本节物理词汇：

Kinematics 运动学
Reference Frames 参照系
Displacement 位移
Instantaneous Velocity 瞬时速度
Acceleration 加速度
Freely Falling Objects 自由落体
Integral Calculus 积分
Numerical Integration 数值积分
Average Velocity 平均速度
Derivative 导数
Variable Acceleration 变加速度
Constant Acceleration 匀加速度
Speedometer 速度计
Gravitational Acceleration 重力加速度
Position-Time Graph 位置-时间图
Velocity-Time Graph 速度-时间图
Area Under Curve 曲线下面积

【公式】

1. Displacement（位移公式）

$\Delta x = x_2 - x_1$

2. Average Velocity（平均速度公式）

$\text{Average Velocity} = \frac{\Delta x}{\Delta t}$

3. Instantaneous Velocity（瞬时速度公式）

$v = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$

4. Average Acceleration（平均加速度公式）

Average Acceleration$=\frac{\Delta v}{\Delta t}$

5. Instantaneous Acceleration（瞬时加速度公式）

$a = \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt} = \frac{d^2x}{dt^2}$

6. Constant Acceleration Motion Equations（匀加速运动方程）

速度方程：$v = v_0 + at$

位移方程：$x = x_0 + v_0 + \frac{1}{2}at^2$

速度-位移关系：$v^2 = v_0{}^2 + 2a(x - x_0)$

7. Freely Falling Objects（自由落体公式）

重力加速度：$g = 9.80\ m/s^2$

位移方程：$y = y_0 + v_0 t - \frac{1}{2}gt^2$

8. Integration for Variable Acceleration（变加速度积分公式）

速度积分：$v = v_0 + \int_0^t a(t)dt$

位移积分：$x = x_0 + \int_0^t v(t)dt$

9. Graphical Analysis（图形分析公式）

位移为速度-时间曲线下面积：$\Delta x = x_0 + \int_{t_1}^{t_2} v(t)dt$

速度为加速度-时间曲线下面积：$\Delta v = x_0 + \int_{t_1}^{t_2} a(t)dt$

【Example 1】

A stone1 freely falls from the edge of an 80-meter-high cliff (with an initial velocity of 0), while another stone2 is vertically thrown upward from the ground with an initial velocity of 20 m/s. Ignoring air resistance, the acceleration due to gravity is g=9.80 m/s^2 (the two stones do not collide).

1.When are the two stones at the same height?

2.The velocities of the two stones at the same height.

3.The time difference when the two stones hit the ground.

【辅助理解】

依据题目信息画出图像，连列等式求解。在竖直上抛运动中，初末位置相同的情况下，若忽略空气阻力且重力加速度恒定，物体的上升阶段与下落阶段具有时间对称，因此第三问求向上抛的石子落地时间时，只需要求石子速度从 20 变化为 0 的时间，再乘以 2 就是落地总时间。

【Solution】

1.Let the time taken for the two stones to fall to the same height is t_1，and the initial velocity of Stone 2 is v_0 .

$$\begin{cases} x_1 = \frac{1}{2}g{t_1}^2 \\ x_2 = v_0 t_1 - \frac{1}{2}g{t_1}^2 \\ x_1 + x_2 = 80 \end{cases}$$

The solution is t_1 = 4 m/s

2. The speed of Stone 1 is：

$\mathbf{v_1 = g t_1} = 9.8 \times 4 = 39.2\mathrm{m/s}$

The speed of Stone 2 is：

$\mathbf{v_2 = v_0 - g t_1} = 20 - 9.8 \times 4 = -19.2\mathrm{m/s}$ （Direction is vertically downward）

3. Let the landing time of Stone 1 be t_3

Then， $\mathbf{H = \frac{1}{2} g {t_3}^2}$

The solution is t_3 ≈ 4.04 m/s

Let the half of the landing time of Stone 2 be t_4，

$0 = v_0 - gt_4 = 20 - 9.8 \times t_4$，

The solution is t_4 ≈ 2.04 m/s，

$\Delta t = 2t_4 - t_3 = 2 \times 2.04 - 4.04 = 0.04\mathrm{s}$

二维或三维空间中的运动学；向量

本节物理词汇：

Components 分量
Unit Vectors 单位矢量
Projectile Motion 抛体运动
Relative Velocity 相对速度
Parallelogram Method 平行四边形法
Tail-to-Tip Method 首尾相接法
Instantaneous Acceleration 瞬时加速度
Gravitational Acceleration 重力加速度
Acceleration Vector 加速度矢量
Trajectory 轨迹
Kinematic Equations 运动学方程
Reference Frame 参照系
Resultant Vector 合矢量

【公式】

1. 位移矢量公式 (Displacement Vector Formula)

$$\Delta\vec{r} = \vec{r_2} - \vec{r_1}$$

(方向由初位置指向末位置)

2. 瞬时速度公式 (Instantaneous Velocity Formula)

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k}$$

3. 瞬时加速度公式 (Instantaneous Acceleration Formula)

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{dv_x}{dt}\hat{i} + \frac{dv_y}{dt}\hat{j} + \frac{dv_z}{dt}\hat{k}$$

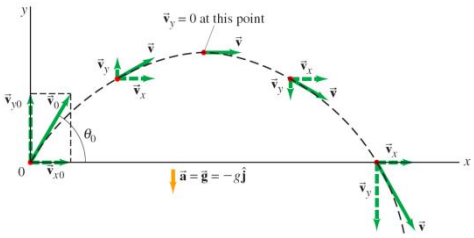
(方向与速度变化方向一致)

【基本思想】

1. 平抛运动

$$\begin{cases} \text{水平：匀速直线运动} \\ \text{竖直：自由落体运动} \end{cases}$$

斜抛运动



求水平面到最高点 h：$h = \frac{1}{2}at^2$

射程 $x = v_x \bullet 2t = v_0 \cos\theta \bullet \frac{2v_0 \sin\theta}{g} = \frac{2{v_0}^2 \sin\theta \cos\theta}{g}$

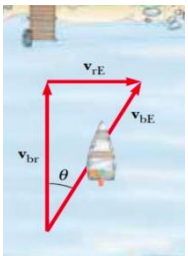
2. 小船过河

（1）最短过河时间：船头垂直于河岸，即$v_{船}$垂直于河岸，但因为有$v_{水}$，所以实际过河速度不垂直于河岸。

（2）过河航程最短：$v_{船}$和$v_{水}$的矢量合速度垂直于河岸。

（3）当$v_{船} < v_{水}$时，不能垂直于河岸过河，因为此时不能组成以$v_{船}$为斜边的矢量三角形。

【Example 1】
A small boat crosses a river with a width of d = 180m. The current speed is v_1= 2.5m/s and the boat's speed relative to the water surface is v_2= 5m/s. Please find:
1. If the boat needs to cross the river in the shortest time, How long does it take? What is the displacement?
2. If the boat needs to cover the shortest distance, in which direction should the boat head?How long does it take?



【辅助理解】

（1）最短过河时间：船头垂直于河岸，即v_{br}垂直于河岸
（2）过河航程最短：v_{br}和v_{rE}的矢量合速度垂直于河岸。

【Solution】
1.According to the question, the bow of the boat should be perpendicular to the direction of the riverbank.

$$t_1 = \frac{d}{v_{br}} = \frac{180}{5} = 36s$$

$$v_{bE} = \sqrt{{v_{br}}^2 + {v_{rE}}^2} = \sqrt{5^2 + 2.5^2} = \frac{5\sqrt{5}}{2} m/s$$

$$x = v_{bE} t_1 = \frac{5\sqrt{5}}{2} \times 36 = 90\sqrt{5} m$$

2.$v_{br} > v_{rE}$， the shortest voyage across the river is equal to the river width. The actual speed of the boat v_{bE} is perpendicular to the riverbank. At this time, the bow of the boat is inclined upwards. Suppose the angle between the bow of the boat and the riverbank is θ, then the angle θ satisfies

$$\cos\theta = \frac{v_{rE}}{v_{br}} = \frac{2.5}{5} = 0.5$$

$$\theta = 60°$$

Hence， When the bow of the ship forms a 60-degree angle with the riverbank, the voyage is the shortest.

$$v_{bE} = v_{br} \sin\theta = \frac{5\sqrt{3}}{2} m/s$$

$$t = \frac{d}{v_{bE}} = 24\sqrt{3} s$$

牛顿运动定律

本节物理词汇：

Inertia 惯性

Inertial reference frame 惯性参考系

Proportional to 正比于

【基本思想】

1. Newton's first law (law of inertia) An object remains at rest or in uniform motion in a straight line without the action of a net force. The law holds in an inertial frame of reference, but not in a non-inertial frame of reference (such as an accelerating or rotating frame).

牛顿第一定律（惯性定律） 物体在无净外力作用下保持静止或匀速直线运动。惯性参考系中定律成立，非惯性系（如加速或旋转参考系）不适用。

2. Newton's second law acceleration is proportional to the net force and inversely proportional to mass: $\sum \vec{F} = m\vec{a}$. Force changes the state of motion (speed or direction) of an object.

牛顿第二定律 加速度与净外力成正比，与质量成反比：$\sum \vec{F} = m\vec{a}$ 。力改变物体的运动状态（速度大小或方向）。

3. Newton's third law The magnitude of the action force is equal to that of the reaction force, and the direction is opposite, acting on different objects: $\vec{F}_{AB} = -\vec{F}_{BA}$.

牛顿第三定律 作用力与反作用力大小相等，方向相反，作用在不同物体上：$\vec{F}_{AB} = -\vec{F}_{BA}$ 。

重力 法向力 摩擦力

本节物理词汇：

Subscript 下标的

Normal force 法向力

Friction 摩擦力

Kinetic friction 动摩擦

Static friction 静摩擦

Free-body diagram 受力分析图

Coordinate system 坐标系统

【基本思想】

1. Gravity and normal force Gravity: $\vec{F}_G = m\vec{g}$, in the vertical downward direction. Normal force: perpendicular to the contact surface, balancing gravity or other external forces.

重力和法向力 重力：$\vec{F}_G = m\vec{g}$，方向竖直向下。法向力：垂直于接触面，平衡重力或其他外力。

2. Friction static friction: $F_{fr} \le \mu_s F_N$ to prevent relative sliding. Dynamic friction: $F_{fr} = \mu_k F_N$ to act on the sliding object.

摩擦力 静摩擦力：$F_{fr} \le \mu_s F_N$，防止相对滑动。动摩擦力：$F_{fr} = \mu_k F_N$，作用于滑动物体。

3. The solution steps are to draw the force analysis diagram, decompose the force, apply Newton's law to set up equations, and solve.

解题步骤 画受力分析图，分解力，应用牛顿定律列方程，求解。

【Example 1】

A block of wood with a mass of 5kg is placed on a horizontal table. The coefficient of static friction is $\mu_s = 0.4$ and the coefficient of dynamic friction is $\mu_k = 0.2$.

1. If a horizontal force of 10N is applied, will the block move?

2. If a horizontal force of 25N is applied, will the block move? If so, find the acceleration of the block.

【辅助理解】

1. 先计算最大净摩擦力 $F_{fr,\max} = \mu_s F_N$，比较外力是否超过它。

2. 若移动，计算动摩擦力 $F_{fr} = \mu_k F_N$，再用牛顿第二定律求加速度。

【Solution】

1. Calculate the normal force F_N and maximum static friction:

$F_N = mg = 5 \times 9.8 = 49N$

$F_{fr,\max} = \mu_s F_N = 0.4 \times 49 = 19.6N$

$10N < 19.6N$ The blocks don't move.

2. If a horizontal force of 25N is applied,

$25N > 19.6N$ The blocks began to slide.

Kinetic force of friction $F_{fr} = \mu_k F_N = 0.2 \times 49 = 9.8N$

Net external forces $F_{net} = 25 - 9.8 = 15.2N$

Acceleration $a = \frac{F_{net}}{m} = \frac{15.2}{5} = 3.04m/s^2$

圆周运动

本节物理词汇：

Uniform circular motion 匀速圆周运动

Radius 半径

Tangent 切线（的）

Tangential 切向的

Normal 法向的

Centripetal 向心（力）的

Banked turn 倾斜转弯

Radial 径向的

【基本思想】

1. Uniform circular motion The velocity direction is along the tangent, and the acceleration points to the center of the circle (centripetal acceleration): $a_R = \frac{v^2}{r}$.

The centripetal force is provided by the net external force: $\sum F_R = m\frac{v^2}{r}$.

匀速圆周运动 速度方向沿切线，加速度指向圆心（向心加速度）：$a_R = \frac{v^2}{r}$。向心力由净外力提供：$\sum F_R = m\frac{v^2}{r}$。

2. Tilted turning and non-tilted turning Non-tilted road: Friction provides the centripetal force. Tilted road: The horizontal component of the normal force provides the centripetal force, reducing the dependence on friction.

倾斜转弯与无倾斜转弯 无倾斜路面：摩擦力提供向心力。倾斜路面：法向力的水平分量提供向心力，减少对摩擦力的依赖。

3. The total acceleration of non-uniform circular motion is divided into tangential component (change the magnitude of velocity) and normal component (change the direction of velocity).

非匀速圆周运动 总加速度分为切向分量（改变速度大小）和法向分量（改变速

度方向）。

4. Dynamical applications The problem of conical pendulum, car turning and so on need to be solved by combining centripetal force and force analysis.

动力学应用 圆锥摆，汽车转弯等问题需结合向心力和受力分析求解。

【Example 1】

A car of 1000kg is traveling at a speed $20m/s$ on a horizontal curve with a radius of 50m.

1. If the road surface is not inclined, find the required minimum static friction coefficient μ_s to prevent slipping.

2. If the road is inclined, find the required friction at this time. (Take μ_s which is from problem 1)

【辅助理解】

1. 当汽车在水平路面上转弯时，向心力必须由静摩擦力提供。

2. 当路面倾斜时，重力的分量会提供部分向心力，因此所需的摩擦力会减少，我们需要分析力的分解。

【Solution】

1. The formula for centripetal force F_c is: $F_c = \frac{mv^2}{r}$

$m = 1000kg$

$v = 20m/s$

$r = 50m$

Substitute the value: $F_c = 8000N$

Static frictional force $f_s = \mu_s N$

$N = mg = 1000 \times 9.8 = 9800N$

To prevent slipping, we need:

$f_s \geq F_c \Rightarrow \mu_s N \geq 8000$

Thus： $\mu_s \geq \frac{8000}{9800} \approx 0.816$

The minimum static friction coefficient required is 0.816.

2. Establish a coordinate system, set the slope Angle as $\theta = 30^\circ$, along the slope upward is the positive direction of x, perpendicular to the slope upward is the positive direction of y

Force decomposition, y direction force balance: $N = 4900\sqrt{3}N \approx 8487N$

Centripetal force $F_c = 8000N$

$F_c = f + mg\sin\theta \Rightarrow f = F_c - mg\sin\theta$

$f = 3100N$, The direction is up the slope.

Check whether the friction is reasonable, and the maximum static friction is: $f_{\max} = \mu_s N$

$f_{\max} = 0.816 \times 8487 \approx 6925N > f = 3100N$,is rational.

The required friction force is 3100N.

【Example 2】

A small ball is suspended by a rope $L = 1m$ and moves in uniform circular motion on the horizontal plane. The rope is at $\theta = 30^\circ$ to the vertical direction.

1. Find the linear velocity of the ball v .
2. How will the ball move if the rope suddenly breaks? (Neglect air resistance)

【辅助理解】

1. 受力分析，向心力公式，平衡方程。
2. 断裂瞬间的运动状态，后续运动轨迹。

【Solution】

1. Vertical direction: $T\cos\theta = mg$

Horizontal direction: $T\sin\theta = m\frac{v^2}{r} = m\frac{v^2}{L\sin\theta}$

$$v^2 = gL\frac{\sin^2\theta}{\cos\theta} \Rightarrow v = \sqrt{gL\frac{\sin^2\theta}{\cos\theta}}$$

$g = 9.8m/s^2$， $L = 1m$， $\theta = 30^\circ$。

$$v \approx \sqrt{9.8\times 0.2887} \approx 1.682m/s$$

2. When the rope breaks, the ball is moving along the tangent of the circle at a speed.

The height of the ball at this time is $h = L - L\cos\theta$

$$h \approx 1\times(1-0.866) \approx 0.134m$$

$$x(t) = vt$$

$$y(t) = h - \frac{1}{2}gt^2$$

$$y(t) = 0 \Rightarrow t = \sqrt{\frac{2h}{g}} \approx 0.165s$$

Level flight distance: $x = vt \approx 1.682\times 0.165 \approx 0.278m$

速度相关力

【基本思想】

1. Speed related forces (such as air resistance) Resistance is related to speed, and at terminal speed resistance is balanced with gravity: $v_T = \frac{mg}{b}$.

速度相关力（如空气阻力） 阻力与速度相关，终端速度时阻力与重力平衡：$v_T = \frac{mg}{b}$。

【Example 1】

一个雨滴从静止开始下落，受到空气阻力 $F_{drag} = -bv$，其中 $b = 0.2kg/s$，雨滴质量 $m = 0.01kg$。

1. 求雨滴的终端速度 v_T。

2. 求雨滴速度随时间变化的表达式 $v(t)$。

3. 计算 $t = 1s$ 时的速度（取 $g = 9.8m/s^2$）

【Solution】

1. 重力：$F_g = mg$（向下） 空气阻力：$F_{drag} = -bv$（向上）

在终端速度时，合力为零：$F_g + F_{drag} = 0 \Rightarrow v_T = \frac{mg}{b}$

代入数值：$v_T = 0.49m/s$

2. 牛顿第二定律：$m\frac{dv}{dt} = mg - bv \Rightarrow \frac{dv}{dt} + \frac{b}{m}v = g$

积分因子法：积分因子：$\mu(t) = e^{\frac{b}{m}t}$

乘以积分因子并积分：

$$e^{\frac{b}{m}t}\frac{dv}{dt} + \frac{b}{m}e^{\frac{b}{m}t}v = ge^{\frac{b}{m}t} \Rightarrow \frac{d}{dt}(ve^{\frac{b}{m}t}) = ge^{\frac{b}{m}t} \Rightarrow ve^{\frac{b}{m}t} = \int ge^{\frac{b}{m}t}dt = \frac{gm}{b}e^{\frac{b}{m}t} + C$$

$$v(t) = \frac{gm}{b} + Ce^{-\frac{b}{m}t}$$

代入数值：$t = 0$，$v = 0$，$v_T = 0.49m/s$

$$v(t) = 0.49(1 - e^{-20t})m/s$$

$$v(1) = 0.49(1 - e^{-20\times 1}) \approx 0.49m/s$$

重力

本节物理词汇：

Gravitation 重力
Tangential speed 切向速度
Orbit 轨道
Planetary motion 行星运动
Weightlessness 失重
Elliptical orbit 椭圆轨道

【公式】

1. Newton's Law of Universal Gravitation（万有引力公式）：

$$F = G\frac{m_1 m_2}{r^2}$$

2. Kepler's 1st Law（开普勒第一定律，也叫轨道定律）:
The orbit of each planet is an ellipse, with the Sun at one focus.
所有行星都沿各自的椭圆轨道运动，太阳位于椭圆的一个焦点上。

3. Kepler's 2nd Law（开普勒第二定律，也叫面积定律）：
An imaginary line drawn from each planet to the Sun sweeps out equal areas in equal times.
太阳系中太阳和运动中的行星的连线 （矢径）在相等的时间内扫过相等的面积。

4. Kepler's 3rd Law（开普勒第三定律, 也叫行星运动规律）：
The square of a planet's orbital period is proportional to the cube of its mean distance from the Sun.
绕以太阳为焦点的椭圆轨道运行的所有行星，其各自椭圆轨道半长轴的立方与周期的平方之比是一个常量。
For any two planets in the Solar System, we can write:

$$\left(\frac{T_1}{T_2}\right)^2 = \left(\frac{s_1}{s_2}\right)^3$$

【Example 1】

A planet of mass m is a distance d from Earth. Another planet of mass 2m is a distance 2d from Earth. And the line connecting the two planets to Earth is perpendicular. Calculate the direction of the total gravitation force on Earth.

【辅助理解】

依据题目信息画出图像，依据万有引力公式对受力物体 （地球）进行分析，之后依据平行四边形准则求出方向。

【Solution】

$$F_1 = \frac{GM_e m}{d^2}$$
$$F_2 = \frac{GM_e 2m}{(2d)^2}$$
$$\tan\theta = \frac{F_1}{F_2} = \frac{1}{2}$$

θ is the angle between the total gravitation and the line of the earth and the planet which mass is m.

【Example 2】

Estimate the acceleration due to gravity at the surface of Europa (one of the moons of Jupiter) given that its mass is 4.9×10^{22} kg and making the assumption that its mass per unit（密度） volume is the same as Earth's. $G=6.67\times10^{-11}$ N·m^2 / kg^2, $M_{Ea}=5.97\times10^{24}$kg, $R_{Ea}=6.371\times10^{6}$m.

【辅助理解】

首先明确重力加速度公式，知道缺少什么条件。已知两星球密度相同，依据星球密度公式，可求出两星球质量与半径的比。故据上述条件可求出未知星球的重力加速度。

【Solution】

From the density formula（密度公式）:

$$\rho = \frac{M}{V} = \frac{M}{\frac{4}{3}\pi R^3}$$
$$\rho_E = \rho_{Ea}$$
$$\frac{M_E}{\frac{4}{3}\pi R^3{}_E} = \frac{M_{Ea}}{\frac{4}{3}\pi R^3{}_{Ea}}$$
$$\frac{M_E}{R^3{}_E} = \frac{M_{Ea}}{R^3{}_{Ea}}$$

Solve for the radius R_E of Europa:

$$R_E = R_{Ea}\sqrt[3]{\frac{M_E}{M_{Ea}}} \approx 0.2R_E = 1.2742\times10^6\mathrm{m}$$

Calculate the gravitational acceleration gE on the surface of Europa:

$$g_E = \frac{GM}{R^2} \approx 2.01\ \mathrm{m/s^2}$$

So the gravitational acceleration on the surface of Europa is approximately 2.01m/s^2.

运用 kinetic enegy

本节物理词汇：

Kinetic energy 动能
Translational kinetic energy 平动动能
Joules 焦耳
Work-energy principle 功能原理
Spring constant 弹性系数
Uncompressed length 弹簧原长

【基本思想】

1. 动能 K (kinetic energy) = $1/2mv^2$

其中 m 为物体质量（mess）, v 为物体运动速度（velocity）。动能是标量。

推导过程：W_{net}（合外力做功）= $F_{net}\,d = m\,a_{net}\,d$

又因为 $2ad=v_2^2 - v_1^2$, $a= (v_2^2 - v_1^2)/2d$

所以 $W_{net} = 1/2mv_2^2 - 1/2mv_1^2$

定义动能 K (kinetic energy) = $1/2mv^2$（可以看出同一物体其动能与速度的平方成正比）

2. Work-energy principle 功能原理

$W_{net} = \Delta K = K_2 - K_1 = 1/2mv_2^2 - 1/2mv_1^2$

The net work done on an object is equal to the change in the object's kinetic energy. 合外力做功等于动能变化量。

合外力做正功，动能增加；合外力做负功，动能减小。

3. 动能单位：焦耳（J）。与功的单位一致。

4. 概念辨析：动量 p=mv 是矢量，其方向与速度方向一致；动能 $K = 1/2mv^2$ 是标量，不受速度方向影响，永远不可能是负数。

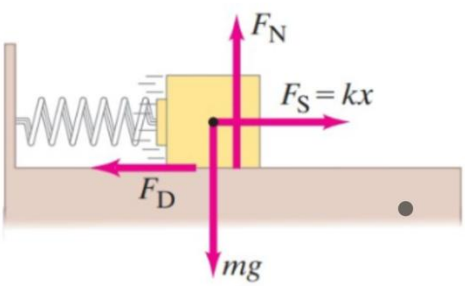
【Example 1】

A horizontal spring has spring constant k = 360 N/m

(a) How much work is required to compress it from its uncompressed length x =0 to x = 11.0 cm?

(b) if a 1.85-kg block is placed against this compressed spring and the spring is released, what will be the speed of the block when it separates from the spring at x = 0? Ignore friction.

(c) repeat part (b) but assu,e that the block is moving on a table as in figure 7-1 and that some kind of constant drag force F_D = 7.0 N is acting to slow it down, such as friction (or perhaps your finger).



【辅助理解】

(a) 计算压缩弹簧所需的功；

弹簧压缩/拉伸时存储弹性势能，做功公式：$W_{弹} = 1/2\,k\,x^2$ （x 为形变量，与平衡位置的距离）

功的本质是能量传递，此处外力对弹簧做功转化为弹性势能。注意单位换算（cm→m）

(b)若水平面光滑（无摩擦），求弹簧释放后木块脱离弹簧时的速度；

光滑水平面无能量损耗，弹簧的弹性势能完全转化为木块的动能：

$$W_{弹} = K = 1/2\ m\ v^2$$

(c) 若存在水平阻力 F_D，求木块脱离弹簧时的速度。

阻力为非保守力，做功会消耗能量（转化为热能），需用功能原理：阻力消耗部分能量，需用总能量减去阻力做功，再求剩余动能对应的速度，即：弹性势能 + 阻力做的负功 = 木块动能。注意，阻力方向与位移相反，做功为负（计算时取 $-F_D x$ ）

公式表达：$1/2\ k\ x^2 - F_D x = 1/2\ m\ v^2$

弹性势能做功公式：$W_{弹} = 1/2\ k\ x^2$

弹力公式：$F_{弹} = k\ x$

【Solution】

（a） $W_{弹} = 1/2\ k\ x^2 = 1/2(360\ N/m)\ (0.110m)^2 = 2.18\ J$

（b） $W_{弹} = K = 1/2\ m\ v^2$, $2.18\ J = 1/2\ m\ v^2$, $v = 1.54\ m/s$

（c） $1/2\ k\ x^2 - F_D x = 1/2\ m\ v^2$,

$- F_D x = -(7.0\ N)\ (0.110m) = -0.77\ J$, $v = 1.23\ m/s$

【Example 2】

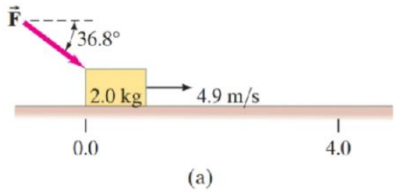
A 2.0-kg block slides across a rough surface with a constant coefficient of kinetic friction of 0.50. The block starts at x=0 with an initial velocity of 4.9 m/s. Pushing the block is a force directed at 36.8 below the horizontal and whose magnitude increases with position.

(a)What is the initial kinetic energy of the block?

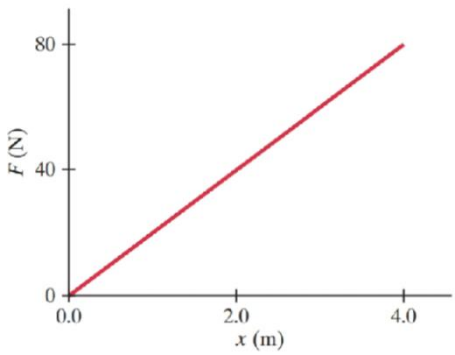
(b)How much work is done by the applied force in the first 4.0m?

(c)What is the magnitude of the friction force at 0.0m, and at 4.0m?

(d)What is the speed of the block at 4.0m?



(a)



(b)

【辅助理解】

(a) Calculate the initial kinetic - energy of the block:

$K = 1/2\ m\ v^2$

Given that the mass of the block m = 2.0 kg and the initial velocity v_0=4.9 m/s.

Substitute the values into the formula: $K_0 = 1/2mv_0^2 = 1/2(2.0\ kg)\ (4.9\ m/s)^2 = 24.01\ J$.

(b) Calculate the work - done by the applied force in the first 4.0 m:
The force F is a linear function of x. The area under the F - x graph from x = 0 to x = 4.0 m is the area of a triangle. The formula for the area of a triangle is A = 1/2 bh.
Here, the base b = 4.0 m and the height h = 80 N. So, the work - done by the applied force W_F = 1/2(4.0 m) (80 N) = 160 J.

(c) Calculate the magnitude of the friction force:
The normal force N on the block is given by N = mg + $F\sin\theta$,
m = 2.0 kg, g = 9.8 m/s^2, F is the applied force,
At x = 0, the frictional force is f_0 = (0.50) (19.6 N) = 9.8 N.
At x = 4.0 m, F = 80 N. Then f_4 = (0.50) (67.6 N) =33.8 N.

(d) Calculate the speed of the block at x = 4.0\ m using the work - energy theorem: The work - energy theorem states that

$$W_{net} = \Delta K = K_2 - K_1 = 1/2mv_2^2 - 1/2mv_1^2$$

The work done by the friction force W_F is the area under the f - x graph.
Since f is linear, W_f=1/2(f_0 + f_4) d, where f_0 = 9.8 N, f_4 = 33.8 N, and d = 4.0 m.
So, W_f=1/2(f_0 + f_4) d = 1/2(9.8 N + 33.8 N) (4.0 m) = 87.2 J.
the work - done by the applied force W_F = 1/2(4.0 m) (80 N) = 160 J
Then the net work done on the block is W_{net}= W_F - W_f =160 J-87.2 J = 72.8 J.
Since W_{net}= K_f – K_0 and K_0 = 24.01 J, then K_f = 96.81 J.
K_f = 1/2 m v_f^2, then v = 9.8 m/s.

【Solution】
(a) The initial kinetic energy of the block is 24.01 J.
(b) The work done by the applied force in the first 4.0 m is 160 J.
(c) The magnitude of the friction force at x = 0 is 9.8 N, at x = 4.0 m is 33.8N.
(d)The speed of the block at x = 4.0 m is approximately 9.8 m/s.

建立能量守恒关系式

本节物理词汇：

conservative force 保守力
nonconservative force d 非保守力
Gravitational potential energy 重力势能
Elastic potential energy 弹性势能
Conserved quantity 守恒量
Total potential energy 总势能
Total mechanical energy 总机械能
Conservation of mechanical energy 机械能守恒
Dissipating energy 耗散力
Gravitational Acceleration 重力加速度
Thermal energy 热能
Internal energy 内能
Escape velocity 逃逸速度
Power 功率
Horsepower 马力

【基本思想】

1. 能量既不能凭空产生，也不能凭空消失，只能从一种形式转化为另一种形式。
2. 机械能：动能，重力势能，弹性势能
3. 机械能守恒的条件：没有摩擦力或其他非保守力做功
4. 机械能守恒定律： $K_1 + U_1 = K_2 + U_2$
5. 能量守恒定律： $K_1 + U_1 = K_2 + U_2 + W_{非}$
6. 功率：单位时间转换的能量
   $P = dE/dt$

【解题步骤】

1. 作图，呈现物体运动情况
2. 瞄准目标物理量，选择运动过程的起点（初始位置）、终点（最终位置），这两个位置的选择对与应用能量守恒方程至关重要。
3. 涉及高度变化：选择合适的重力势能参考面
   涉及弹簧形变：注意弹簧原长
4. 如果没有摩擦力或其他非保守力做功，则可以运用机械能守恒定律：
   $$K_1 + U_1 = K_2 + U_2$$
5. 能量守恒定律：
   若存在非保守力做功，$K_1 + U_1 = K_2 + U_2 + W_{非}$
   根据过程中机械能增加还是减少来确定 $W_{非}$的正负性。
   若非保守力为恒力（恒定不变），$W_{非} = F_{非}d$
   若非保守力不恒定，采用积分。
6. 对所列出的方程求解。
7. 注意单位统一。

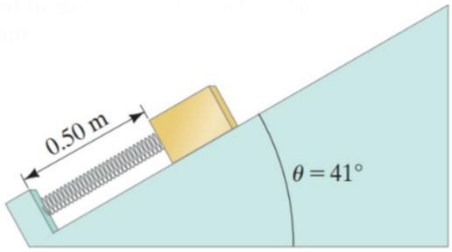
【Example 1】

A spring (k=75 N/m) has an equilibrium length of 1.00m. The spring is compressed to a length of 0.50 m and a mass of 2.0 kg is placed at its free end on a frictionless slope which makes an angle of 41 with respect to the horizontal. The spring is then released.

(a) If the mass is not attached to the spring, how far up the slope will the mass move before coming to rest?

(b) If the mass is attached to the spring, how far up the slope will the mass move before coming to rest?

(c) Now the incline has a coefficient of kinetic friction μ_k. If the block, attached to the spring, is observed to stop just as it reaches the spring's equilibrium position, what is the coefficient of friction μ_k?



【辅助理解】

一根弹簧（k=75 N/m）的平衡长度为 1.00 米。 将弹簧压缩至 0.50 米长，并在其自由端放置一个质量为 2.0 千克的物体，该物体位于一个与水平面成 41 度角的无摩擦斜坡上。然后释放弹簧。

(a) 如果质量不固定在弹簧上，质量在斜坡上移动多远才会静止？
即物块不与弹簧连接时，沿斜面上升的最大距离。

核心思路：弹簧的弹性势能完全转化为物块的重力势能。

弹簧压缩量 $\Delta x = 1.00\ m - 0.50\ m = 0.50\ m$ ，

弹性势能 $U_{弹} = 1/2\ k(\Delta x)^2 = 9.375\ J$.

该过程中，弹性势能转化为重力势能 $U_{重} = mgh$ ，

由 $U_{弹} = U_{重}$ 得：

$1/2\ k(\Delta x)^2 = mgh = mgd\sin\theta$ (d 是斜面方向上上升的距离)

(b) 如果质量连在弹簧上，质量在斜面上移动多远才会静止？
即物块与弹簧连接时，沿斜面上升的最大距离。

核心思路：弹簧的弹性势能转化为物块的重力势能和弹簧的新弹性势能（因弹簧被拉伸）。

设物块上升距离为 d

弹簧最终伸长量为 d - 0.50 m （原压缩量 0.50 m），

末态弹性势能 $U'_{弹} = 1/2\ k\ (d - 0.50m)^2$.

重力势能 $U_{重} = mgd\sin\theta$

能量守恒：初始弹性势能 = 末态弹性势能 + 重力势能

$1/2\ k\ (0.50m)^2. = 1/2\ k\ (d - 0.50m)^2 + mgd\sin\theta$

展开化简后得方程并求解

(c) 现在斜面的动摩擦系数为 μ_k。如果观察到弹簧上的木块在到达弹簧的平衡位置时停止，那么摩擦系数 μ_k 是多少？
求物块停在弹簧原长位置（ 1.00 m ）时的动摩擦因数

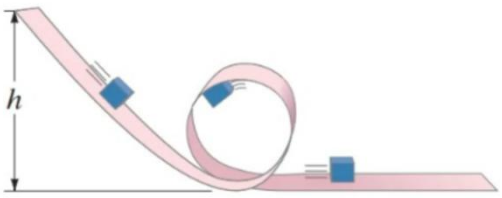
核心思路：弹性势能用于克服摩擦做功和增加重力势能。
物块移动距离：从压缩位置到原长位置，移动 d = 0.50m
摩擦力做功：
摩擦力 $f = \mu_k N = \mu_k mg\cos\theta$
做功 $W_f = f d = \mu_k mgd\cos\theta$
能量守恒：弹性势能 = 摩擦做功 + 重力势能

$$1/2\, k(\Delta x)^2 = \mu_k mgd\cos\theta + mgd\sin\theta$$

代入数据求解

【Solution】
(a) d= 0.73m
无连接时，弹性势能全转重力势能
(b) d = 0.66m
有连接时，弹性势能分给重力势能和弹簧伸长的弹性势能
(c) d = 0.40m
有摩擦时，需考虑摩擦消耗的能量

【Example 2】
The small mass m sliding without friction along the looped track is to remain on the track at all times, even at the very top of the loop of radius r.
(a) In terms of the given quantities, determine the minimum release height h.
(b) If the actual release height is 2h, calculate the normal force exerted by the track at the bottom of the loop, then
(c) by the track at the top of the loop, and
(d) by the track after the block exits the loop onto the flat section.



【辅助理解】

一个质量为 m 的物块从高度 h 处沿无摩擦轨道下滑，进入半径为 r 的竖直圆轨道。求：
(a) 物块能通过圆轨道顶端的最小释放高度 h；
思路：临界条件为顶端，该点处仅重力提供向心力，结合机械能守恒求最小高度
(b) 若释放高度 2h ，物块到达圆轨道底部时轨道对它的支持力；
(c) 若释放高度为 2 ，物块到达圆轨道顶端时轨道对它的支持力；
思路：通过机械能守恒求速度，再用牛顿第二定律分析向心力；
(d) 物块滑出圆轨道进入水平段后，轨道对它的支持力。
思路：水平段无竖直加速度，支持力与重力平衡。

总结关键点：各位置的能量转化和受力特点

【Solution】

(a) 已知条件：物块通过圆轨道顶端时，轨道支持力 $N = 0$
顶端临界速度：
由向心力公式 $mg = mv_{顶}^2/r$ 求解
机械能守恒：
释放点势能 = 顶端势能 + 顶端动能
$mgh = mg(2r) + 1/2\ mv_{顶}^2/r$
解得 $h = 2r + 1/2\ r = 5/2\ r$

(b) 底部支持力
底部速度：释放点势能 = 底部动能
$mg(5r) = 1/2\ mv_{底}^2$ ，$v_{底}^2 = 10gr$
受力分析：支持力向上，重力向下，合力提供向心力
$N_{底} - mg = mv_{底}^2/r$
代入 $v_{底}^2 = 10gr$，
得 $N_{底} = mg + 10mg = 11mg$

(c) 顶端支持力
释放点势能 = 顶端势能 + 顶端动能：
$mg(5r) = mg(2r) + 1/2\ mv_{顶}^2$ ，$v_{顶}^2 = 6gr$
受力分析： 支持力向下，重力向下，合力提供向心力
$N_{顶} + mg = mv_{顶}^2/r$
代入，得 $N_{顶} = 5mg$

(d) 水平段支持力：水平段物块竖直方向受力平衡，支持力等于重力
$N = mg$

线性动量

本节物理词汇：

Momentum 动量
Kinetic energy 动能
Elastic collision 弹性碰撞
Inelastic collision 非弹性碰撞
Translational motion 平动

【公式】

1. Definition of Momentum(动量的定义):

$$\vec{p} = m\vec{v}$$

The rate of change of momentum is equal to the net force: $\Sigma\vec{F} = \frac{d\vec{p}}{dt}$

2.Law of Conservation of Momentum（动量守恒定理）:

$$m_A\vec{v}_A + m_B\vec{v}_B = m_A\vec{v}_A' + m_B\vec{v}_B'$$

3.Center of Mass Theorem(质心定理):

$$x_{CM} = \frac{m_A x_A + m_B x_B}{m_A + m_B}$$

$$\vec{r}_{CM} = \frac{1}{M}\int \vec{r}dm$$

4. Center-of-Mass Motion Theorem（质心运动定理）:

$$M\vec{a}_{CM} = \Sigma\vec{F}_{ext}$$

【Example 1】

Two small balls undergo an elastic collision on a frictionless horizontal surface.

（1）Ball A has a mass $m_A = 1kg$ and initial velocity $\vec{v}_A = 3m/s$.
Ball B has a mass $m_B = 1kg$ and initial velocity $\vec{v}_B = -1m/s$.

Find the velocities $\vec{v}_A'$ and $\vec{v}_B'$ of the two balls after the collision.

（2）Ball A has a mass $m_A = 2kg$ and initial velocity $\vec{v}_A = 3m/s$.
Ball B has a mass $m_B = 1kg$ and initial velocity $\vec{v}_B = 0m/s$.

Find the velocities $\vec{v}_A'$ and $\vec{v}_B'$ of the two balls after the collision.

(3) Ball A has a mass $m_A = 2kg$ and initial velocity $\vec{v}_A = 3m/s$.
Ball B has a mass $m_B = 1kg$ and initial velocity $\vec{v}_B = -1m/s$.

【辅助理解】两小球经历弹性碰撞，动能守恒，动量守恒。

【Solution】

（1） $$m_A\vec{v}_A + m_B\vec{v}_B = m_A\vec{v}_A^{'} + m_B\vec{v}_B^{'} \Rightarrow \vec{v}_A + \vec{v}_B = \vec{v}_A^{'} + \vec{v}_B^{'}$$

$$\frac{1}{2}m_A\vec{v}_A{}^2 + \frac{1}{2}m_B\vec{v}_B{}^2 = \frac{1}{2}m_A\vec{v}_A^{'}{}^2 + \frac{1}{2}m_B\vec{v}_B^{'}{}^2 \Rightarrow \vec{v}_A{}^2 + \vec{v}_B{}^2 = \vec{v}_A^{'}{}^2 + \vec{v}_B^{'}{}^2$$

The only solution satisfying both is：

$$\vec{v}_B = \vec{v}_A^{'} = -1\,m/s$$

$$\vec{v}_A = \vec{v}_B^{'} = 3\,m/s$$

（In a head-on elastic collision between two objects of equal mass, their velocities are exchanged post-collision.）

（2） $$m_A\vec{v}_A + m_B\vec{v}_B = m_A\vec{v}_A^{'} + m_B\vec{v}_B^{'}$$

$$\frac{1}{2}m_A\vec{v}_A{}^2 + \frac{1}{2}m_B\vec{v}_B{}^2 = \frac{1}{2}m_A\vec{v}_A^{'}{}^2 + \frac{1}{2}m_B\vec{v}_B^{'}{}^2$$

$$\vec{v}_A^{'} = \frac{m_A - m_B}{m_A + m_B}\vec{v}_A = -1\ m/s$$

$$\vec{v}_B^{'} = \frac{2m_A}{m_A + m_B}\vec{v}_A = 2\ m/s$$

（The above two equations represents the velocity formulas for two objects after an elastic collision when one object is initially at rest.）

（3） $$m_A\vec{v}_A + m_B\vec{v}_B = m_A\vec{v}_A^{'} + m_B\vec{v}_B^{'}$$

$$\frac{1}{2}m_A\vec{v}_A{}^2 + \frac{1}{2}m_B\vec{v}_B{}^2 = \frac{1}{2}m_A\vec{v}_A^{'}{}^2 + \frac{1}{2}m_B\vec{v}_B^{'}{}^2$$

$$\vec{v}_A^{'} = \frac{1}{3}\,m/s$$

$$\vec{v}_B^{'} = \frac{13}{3}\,m/s$$

The third question seems to involve a large amount of calculation, but we have a new way to solve it：

Application of König's Theorem（柯尼希定理） to Elastic Collisions

In the COM reference frame（在质心系中）:

1. Center of mass velocity（质心速度）：$\vec{v}_{CM} = \frac{m_A\vec{v}_A + m_B\vec{v}_B}{m_A + m_B}$
2. Switch to COM Reference Frame（转换到质心参考系）

Initial relative velocities 初始相对速度: $\Delta\vec{v}_A = \vec{v}_A - \vec{v}_{CM}$ $\Delta\vec{v}_B = \vec{v}_B - \vec{v}_{CM}$

Elastic collision property 弹性碰撞性质：$\Delta\vec{v}_A^{'} = -\Delta\vec{v}_A$ $\Delta\vec{v}_B^{'} = -\Delta\vec{v}_B$

3. Transform Back to Lab Frame 转回实验室系

$$\vec{v}_A^{'} = \vec{v}_{CM} + \Delta\vec{v}_A^{'} \qquad \vec{v}_B^{'} = \vec{v}_{CM} + \Delta\vec{v}_B^{'}$$

Back to（3） $$\vec{v}_{CM} = \frac{5}{3}\,m/s$$

$$\Delta\vec{v}_A = -\frac{4}{3}\mathrm{m/s}$$

$$\Delta\vec{v}_B = \frac{8}{3}\mathrm{m/s}$$

$$\vec{v}_A' = \vec{v}_{CM} + \Delta\vec{v}_A' = \frac{5}{3} - \frac{4}{3} = \frac{1}{3}\mathrm{m/s}$$

$$\vec{v}_B' = \vec{v}_{CM} + \Delta\vec{v}_B' = \frac{5}{3} + \frac{8}{3} = \frac{13}{3}\ \mathrm{m/s}$$

【Example 2】

A bullet of mass m is fired horizontally with an initial speed $\mathbf{v}$ into a stationary wooden block of mass $\mathbf{M}$ suspended by a light string. The bullet embeds itself in the block (perfectly inelastic collision), and the block-bullet system swings upward, reaching a maximum height $\mathbf{h}$. Neglect air resistance and assume the string remains taut during the motion.

【辅助理解】

两小球经历非弹性碰撞，动量守恒，能量守恒。

【Solution】

Momentum conservation：

$$mv = (m + M)V$$

$$V = \frac{mv}{m + M}$$

Energy conservation：$\frac{1}{2}(m + M)\left(\frac{mv}{m+M}\right)^2 = (m + M)gh$

$$V = \frac{m + M}{m}\sqrt{2gh}$$

【Example 3】

Three particles are placed at the corners of a right triangle as shown below. The masses and positions are as follows:

Particle A ($m_A = 3\mathrm{kg}$) is located at the origin $(0,0)$.

Particle B ($m_B = 4\mathrm{kg}$) is placed along the x-axis at $(2.0\mathrm{m}, 0)$

Particle C ($m_C = 5\mathrm{kg}$) is placed along the y-axis at $(0, 1.5\ \mathrm{m})$.

【辅助理解】用这两个公式 $X_{cm} = \frac{\Sigma m_i x_i}{\sum m_i}$ $Y_{cm} = \frac{\Sigma m_i y_i}{\sum m_i}$

【Solution】

$$m_A + m_B + m_C = 12\mathrm{kg}$$

$$X_{cm} = \frac{\Sigma m_i x_i}{\sum m_i} = \frac{2}{3}\mathrm{m}$$

$$Y_{cm} = \frac{\Sigma m_i x_i}{\sum m_i} = \frac{5}{8}\mathrm{m}$$

$$\text{Center of mass} = (\frac{2}{3}\mathrm{m},\ \frac{5}{8}\mathrm{m})$$

旋转运动

本节物理词汇：

Angular displacement 角位移
Angular velocity 角速度
Angular acceleration 角加速度
Linear velocity 线速度
Tangential acceleration 切向加速度
Centripetal acceleration 向心加速度
Torque 力矩
Lever arm 力臂
Axis of rotation 旋转轴
Rotational inertia 转动惯量
Parallel-axis theorem 平行轴定理
Perpendicular-axis theorem 垂直轴定理
Rotational kinetic energy 转动动能
Translational kinetic energy 平动动能
Rolling motion 滚动运动

【公式】

1. Angular Kinematics（角运动学）

Angular displacement 角位移：$\theta = l / R$

Average angular velocity 平均角速度：$\overline{\omega} = \Delta\theta / \Delta t$

Instantaneous angular velocity 瞬时角速度：$\omega = d\theta / dt$

Angular acceleration 角加速度：$\alpha = d\omega / dt = \Delta\omega / \Delta t$

2. Linear and Angular Relations（线量与角量关系）

Linear velocity 线速度与角速度：$v = R\omega$

Tangential acceleration 切向加速度：$a_{tan} = R\alpha$

Centripetal acceleration 向心加速度：$a_R = \omega^2 R = v^2 / R$

3. Constant Angular Acceleration（恒角加速度运动）

$$\omega = \omega_0 + \alpha t$$

$$\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$$

$$\omega^2 = \omega_0^2 + 2\alpha\Delta\theta$$

4. Torque（力矩）

$$\tau = F \cdot R \cdot \sin\theta$$

$\sum\tau = I\alpha$ （Rotational Newton's second law 旋转版牛顿第二定律）

5. Moment of Inertia（转动惯量）

Discrete mass 离散质量：$I = \Sigma m_i R_i^2$

Continuous mass 连续质量：$I = \int R^2\, dm$

Parallel-axis theorem 平行轴定理：$I = I_{cm} + Mh^2$

Perpendicular-axis theorem 垂直轴定理（薄板）：$Iz = Ix + Iy$

6. Rotational Kinetic Energy（转动动能）

Pure rotational kinetic energy 纯转动动能：

$$K_{rot} = \frac{1}{2} I\omega^2$$

Total kinetic energy in rolling motion 滚动总动能：

$$K_{total} = \frac{1}{2} Mv^2 + \frac{1}{2} I_{cm}\, \omega^2$$

7. Work and Power（功与功率）

Work done by torque 扭矩做功：$W = \int \tau\, d\theta$

Power 功率：$P = \tau\omega$

8. 常见物体转动惯量

Thin ring 细环（R_0）：$I = MR_0^{\ 2}$

Solid cylinder 实心圆柱：$I = \frac{1}{2} MR_0^{\ 2}$

Thin rod 细杆（center,中心）：$I = \frac{1}{12} Ml^2$

Thin rod 细杆（end,端点）：$I = \frac{1}{3} Ml^2$

Solid sphere 实心球体：$I = \frac{2}{5} Mr_0^{\ 2}$

【Example 1】

A uniform solid disk has a mass $M = 4\ kg$ and radius $R = 0.5\ m$.

1. Calculate the moment of inertia I_{cm} about the central axis.
2. If the disk rotates about a parallel axis at a distance $d = 0.2\ m$ from the center. Calculate the new moment of inertia I
3. A torque $\tau = 8\ N \cdot m$ is applied to the disk about this parallel axis. Determine the angular acceleration α.

【辅助理解】

直接套对公式即可

1.计算中心轴转动惯量：直接套圆盘公式

2.用平行轴定理求新转动惯量： 在原结果上加 Md^2 就是新轴的转动惯量

3.求角加速度：用扭矩除以新转动惯量就得到角加速度

【Solution】

1. Its moment of inertia about the central axis is:

$$I_{cm} = \frac{1}{2} MR^2 = 1/2 \times 4 \times (0.5)^2 = 0.5 kg \cdot m^2$$

2.The **parallel-axis theorem** states:

$$I = I_{cm} + Md^2$$

Given d=0.2 m, substitute the values:

$$I = 0.5 + 4 \times (0.2)^2 = 0.5 + 0.16 = 0.66\,\mathrm{kg \cdot m^2}$$

3. Using the rotational form of Newton's second law:

$$\tau = I\alpha$$

$$\alpha = \frac{\tau}{I} = \frac{8}{0.66} \approx 12.12\mathrm{rad/s^2}$$

【Example 2】

A thin rod of mass $M = 3\,\mathrm{kg}$ and length $L = 2\,\mathrm{m}$ is initially at rest and rotates about an axis through its center.

1. Calculate the moment of inertia I_{cm} about the central axis.
2. A tangential force $F = 6\,\mathrm{N}$ is applied at one end for $t = 2\,\mathrm{s}$. Find the final angular velocity ω.
3. Determine the rotational kinetic energy K_{rot} at this instant.

【辅助理解】

1.算中心转动惯量：细杆公式 $I = \frac{1}{12} Ml^2$ 直接代数字。

2.求角速度：先算扭矩（力×力臂），再除转动惯量得角加速度，最后乘时间得角速度。

3.算转动动能：把转动惯量和角速度代入$K_{rot} = \frac{1}{2} I\omega^2$公式就行

【Solution】

1. For a thin rod rotating about its center:

$$I_{cm} = \frac{1}{12} Ml^2 = \frac{1}{12} \times 3 \times 2^2 = 1\mathrm{kg \cdot m^2}$$

2. First, calculate the torque τ applied at the rod's end:

$$\tau = F \times \frac{L}{2} = 6 \times 1 = 6\,\mathrm{N \cdot m}$$

Next, find the angular acceleration α：

$$\alpha = \frac{\tau}{I_{cm}} = 6\,\mathrm{rad/s^2}$$

Since the rod starts from rest, the final angular velocity after $t = 2\,\mathrm{s}$ is:

$$\omega = \omega 0 + \alpha t = 0 + 6 \times 2 = 12\,\mathrm{rad/s}$$

3. The rotational kinetic energy is given by:

$$K_{rot} = \frac{1}{2} I_{cm}\,\omega^2 = \frac{1}{2} \times 1 \times (12)^2$$

【Example 3】

A uniform solid cylinder with mass $M = 5\,\mathrm{k}$ and radius $R = 0.3\,\mathrm{m}$ rolls without slipping down an inclined plane at angle θ=30°. The plane has length $L = 2\mathrm{m}$ and connects to a rough horizontal surface (coefficient of kinetic friction μ=0.2).

Determine:

1. The moment of inertia I_{cm} of the cylinder;
2. The linear velocity v and angular velocity ω at the bottom of the incline;
3. (Challenge) The time t until pure rolling begins on the horizontal surface, and the final linear velocity v_f.

【辅助理解】

1. 实心圆柱的转动惯量公式
2. 无滑动滚动时，重力势能转化为平动和转动动能
3. 摩擦力使平动减速而转动加速，直到满足 $v = R\omega$ 时变为纯滚动

【Solution】

1. For a uniform solid cylinder, the moment of inertia about its central axis is given by:

$$I_{cm} = \frac{1}{2}MR^2 = \frac{1}{2} \times \mathbf{5} \times (\frac{3}{10})^2 = \frac{9}{40} \quad kg \cdot m^2$$

2. Energy conservation equation 能量守恒方程：

$$Mgh = \frac{1}{2}Mv^2 + \frac{1}{2}I_{cm}\omega^2$$

Using the condition: $\omega = \frac{v}{R} = \frac{10}{3}v$

$$50 = \frac{5}{2}v^2 + \frac{1}{2} \times \frac{9}{40} \times \left(\frac{10v}{3}\right)^2 \Rightarrow v = \frac{2\sqrt{30}}{3} m/s$$

The angular velocity is: $\omega = \frac{10}{3} \times \frac{2\sqrt{30}}{3} = \frac{20\sqrt{30}}{9} rad/s$

3. Step 1: Forces and Accelerations

$$f = \mu Mg = 10N$$
$$a = -\mu g = -2m/s^2$$
$$\alpha = \frac{\tau}{I_{cm}} = \frac{40}{3} rad/s^2$$

Step 2: Time Until Pure Rolling

$$\omega(t) = \omega_0 + \alpha t$$

$$v(t) = v_0 + at$$

Set $v(t) = R\omega(t)$

$$t = \frac{\sqrt{30}}{15} s$$

$$v_f = \frac{8\sqrt{30}}{15} m/s$$

角动量

本节物理词汇：

angular momentum 角动量
conservation of angular momentum 角动量守恒定律
net external torque 净外部转矩

【基本思想】

1. The torque vector: 力矩可以用向量表示，但是力矩向量不是真正意义上的向量，他只有两个正负方向，可以直接求和，符号仅仅代表刚体旋转的方向。

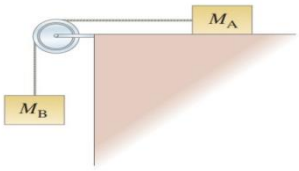
$$\sum \tau = r \times F$$

2. Conservation of angular momentum: 如果净外部转矩为 0，也就是说所有力对应的力矩能互相抵消或者根本没有力矩，系统角动量的和是一个常量。

$$I\omega = I_0\omega_0 = C$$

【Example 1】

Figure below shows two masses connected by a cord passing over a pulley of radius R_0 and moment of inertia I. Mass M_A slides on a frictionless surface and M_B hangs freely. Determine a formula for (a) the angular momentum of the system about the pulley axis, as a function of the speed v of mass M_A or M_B, and (b) the acceleration of the masses. Vector components of the linear acceleration are:



The linear momentum of M_A is $M_A v$, and the perpendicular distance from the pulley axis to the line of motion of M_A is R_0. So the angular momentum of $L_A = M_A R_0 v$. So does L_B.

The pulley is rotating with an angular velocity $\omega = \frac{v}{R_0}$.

The total angular momentum of the system is

$$L = (M_A + M_B)R_0 v + I\frac{v}{R_0}$$

The net torque acting on the system about the pulley axis is

$$\tau = M_B g R_0$$

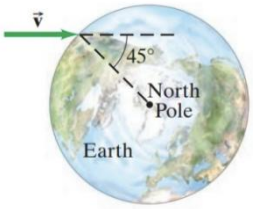
Differentiating L with respect to time t

$$\frac{dL}{dt} = \left[(M_A + M_B)R_0^2 + I\right]\frac{a}{R_0}$$

$$\frac{dL}{dt} = \tau$$

$$a = \frac{M_B g R_0^2}{(M_A + M_B)R_0^2 + I}$$

【Example 2】 Suppose a 5.2×10^{10} kg meteorite struck the Earth at the equator with a speed $v = 2.2 \times 10^4$ m/s, as shown in Fig. and remained stuck. By what factor would this affect the rotational frequency of the Earth ?



Recall that the inertia of solid sphere is $I = \frac{2}{5}MR^2$

The angular velocity of Earth before the impact is $\omega_1 = \frac{2\pi}{1\text{day}}$

The angular momentum of the meteorite is $L_m = mRv\sin 45^\circ$

The initial angular momentum is $L_1 = I_E + L_m$

The new moment of inertia is $I_f = \frac{2}{5}MR^2 + mR^2$

The final angular momentum is $L_f = I_f\omega_2$

According to the law of conservation of angular momentum

$$L_f = L_i$$

$$\frac{2}{5}MR^2\omega_1 + mRv\sin 45^\circ = R^2\left(\frac{2}{5}M + m\right)\omega_2$$

Finally we get $\frac{\omega_2}{\omega_1} = 1 + 7.3 \times 10^{-13}$

振荡

本节物理词汇：

Oscillation/vibration 振荡,振动
Amplitude 振幅
Phase 相位
Simple harmonic motion (SHM)/oscillator (SHO) 简谐运动/振子
Damped harmonic motion 阻尼谐运动
Simple pendulum 单摆
Physical pendulum 物理摆（复摆）
Forced oscillation 受迫振动
Resonance 共振
Periodic 周期的
Equilibrium position 平衡点
Spring constant 弹簧常数（弹性系数）
Periodic driving force 周期性驱动力
Natural frequency 固有频率
Restoring force 恢复力
Massless cord 无质量细绳
Small angle approximation 小角度近似（sinθ ≈ θ）
Damping 阻尼
Resonant peak 共振峰
Sharpness of resonance 共振的锐度
Driven oscillation 驱动振荡

【公式】

1. 胡克定律（Hooke's Law）

$$F = -kx$$

弹簧的回复力与位移成正比，方向指向平衡位置

2. 简谐运动的微分方程（Equation of Simple Harmonic Motion）

$$m\frac{d^2x}{dt^2} + kx = 0$$

$$\frac{d^2x}{dt^2} + \omega^2 x = 0 \quad (\omega = \sqrt{\frac{k}{m}})$$

3. 简谐运动的通解（General Solution of SHM）

$$x(t) = A\cos(\omega t + \varphi)$$

A 为振幅，ω为角频率，φ为初相位

4. 角频率、周期、频率关系（Angular Frequency, Period, Frequency）

$$\omega = 2\pi f = \frac{2\pi}{T} = \sqrt{\frac{k}{m}}$$

$$T = 2\pi\sqrt{\frac{m}{k}}$$

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

5. 速度和加速度公式（Velocity and Acceleration in SHM）

$$v(t) = -A\omega\sin(\omega t + \varphi)$$
$$a(t) = -A\omega^2\cos(\omega t + \varphi) = -\omega^2 x(t)$$

6. 弹簧的势能（Potential Energy of a Spring）

$$U = \frac{1}{2}kx^2$$

7. 简谐振子的总机械能（Total Mechanical Energy of SHM）

$$E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

8. 单摆的周期（Period of a Simple Pendulum）

$$T = 2\pi\sqrt{\frac{l}{g}}$$

l 为摆长，g 为重力加速度（小角度近似）

【Example 1】
A 0.5 kg mass is attached to a spring with a spring constant of 200 N/m. The mass is pulled 0.1 m to the right of equilibrium and released from rest on a frictionless surface.
(1) Find the angular frequency ω.
(2) Find the amplitude A.
(3) Find the maximum speed of the mass.

【辅助理解】
本题考查简谐运动的基本公式，包括角频率、振幅和最大速度的计算。要注意初始条件的应用，以及能量守恒在速度求解中的作用。

【Solution】
(1)

$$\omega = \sqrt{\frac{k}{m}} = 20\mathrm{rad/s}$$

(2) 初始速度为 0，说明此时在最大位移处，A = 0.1 m。

(3)

$$\nu_{max} = \omega A = 20 \times 0.1 = 2.0\ \mathrm{m/s}$$

【Example 2】A simple pendulum has a length of 1.0 m.
(1) Calculate its period T (use $g = 9.8\mathrm{m/s^2}$).
(2) If the length is increased to 4.0 m, what is the new period

【辅助理解】本题考查单摆周期公式的应用。要理解周期与摆长的关系，周期与质量无关，只与长度和重力加速度有关。

【Solution】
(1)

$$T_1 = 2\pi\sqrt{\frac{1.0}{9.8}} = 2.01\mathrm{s}$$

(2)

$$T_2 = 2\pi\sqrt{\frac{4.0}{9.8}} = 4.02\mathrm{s}$$

【Example3】A spring oscillator has amplitude A=0.1 m and spring constant k=50 N/m. At position x=0.06, the speed is v=0.8 m/s.
(1) Find the mass m.
(2) Find the acceleration at this position.
【辅助理解】本题结合了简谐运动的能量守恒和加速度计算。利用总机械能守恒关系求质量，再用加速度公式求指定位置的加速度。

【Solution】
(1) 能量守恒：

$$\frac{1}{2}kA^2 = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

代入数值：

$$\frac{1}{2} \times 50 \times 0.1^2 = \frac{1}{2}m \times (0.8)2 + \frac{1}{2} \times 50 \times 0.06^2$$
$$0.25 = 0.32m + 0.09$$
$$0.16 = 0.32m \Longrightarrow m = 0.5\mathrm{kg}$$

(2) 加速度计算：

$$a = -\frac{k}{m}x = -0.6\ \mathrm{m/s^2}$$

波动

本节物理词汇：

Transverse wave 横波
Longitudinal wave 纵波
Wave velocity 波速
Sinusoidal 正弦曲线
Wave front 波前/波阵面
Reflection 反射
Transmission 透射
Interference 干涉
Diffraction 衍射
Constructive/Destructive interference 相长/相消干涉
Standing wave 驻波
Fundamental frequency 基频
Harmonics 谐波

【公式】

1. 振幅（Amplitude，A）：波的最大位移。

波长（Wavelength，λ）：相邻同相位点间的距离。

频率（Frequency，f）：单位时间内完整周期数。$f = \frac{1}{T}$

波速（Wave velocity，v）：$v = \lambda f = \frac{\lambda}{T}$

相位（φ）：$\varphi = \omega t + \varphi_0$

2. 横波在绳上的波速：$v = \sqrt{\frac{F_T}{\mu}}$ (v 表示速度，F_T表示绳子中的张力)

纵波在介质中的波速：$v=\sqrt{\frac{E}{\rho}}$ or $v = \sqrt{\frac{B}{\rho}}$

3. （1）波的表达式：$D(x) = A\sin\frac{2\pi}{\lambda}x$

辅助理解：$D(x) = A\sin\omega t = A\sin\frac{2\pi}{T}t = A\sin\frac{2\pi}{\frac{\lambda}{v}}t = A\sin\frac{2\pi}{\lambda}x$

（2）经过时间 t 后，波峰已经传播了距离 vt：

$$D(x,t) = A\sin\left[\frac{2\pi}{\lambda}(x - vt)\right]$$

化简：

$$D(x,t) = A\sin(kx - \omega t)$$

$$\omega = 2\pi f\,,\quad k = \frac{2\pi}{\lambda}$$

4. 筑波的波长和频率

$$\lambda_n = \frac{2l}{n}\quad, n = 1,2,3...$$

$$f_n = \frac{v}{\lambda_n} = n\frac{v}{2l} = nf_1\quad, n = 1,2,3...$$

【基本思想】

1. All types of traveling waves transport energy.
波动传递能量，但不传递介质本身。

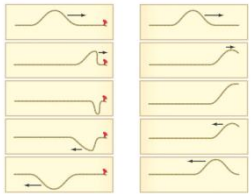
2. Continuous waves start with vibrations, too. If the vibration is SHM, then the wave will be sinusoidal.
连续波由周期性振动产生，若振动为简谐运动（SHM），则波为正弦波。

3. The motion of particles in a wave can be either perpendicular to the wave direction (transverse) or parallel to it (longitudinal).

横波：介质振动方向与波传播方向垂直（如光波、弦上的波）。
纵波：介质振动方向与波传播方向平行（如声波、弹簧压缩波）。
横波在固体、液体表面传播；纵波可在固体、液体、气体中传播。

4. （1）Superposition: The displacement at any point is the vector sum of the displacements of all waves passing through that point at that instant.
（2）Fourier's theorem: Any complex periodic wave can be written as the sum of sinusoidal waves of different amplitudes, frequencies, and phases.
多列波在同一介质中传播时，总位移为各波位移的矢量和。
傅里叶定理：任何复杂周期波可分解为多个正弦波的叠加。

5. （1）A wave hitting an obstacle will be reflected (1), and its reflection will be inverted.
（2）A wave reaching the end of its medium, but where the medium is still free to move, will be reflected (2), and its reflection will be upright.
（1）固定端反射：反射波相位反转（倒置）。
（2）自由端反射：反射波相位不变（直立）。



（1）　（2）

6. A wave encountering a denser medium will be partly reflected and partly transmitted; if the wave speed is less in the denser medium, the wavelength will be shorter.
波从低密度介质进入高密度介质时，部分反射，部分透射，波长变短。

7. 相长干涉：波峰与波峰叠加，振幅增强（相位差为 2nπ）。
相消干涉：波峰与波谷叠加，振幅减弱（相位差为(2n+1)π）。
两列波振幅均为 A，叠加后：
相长干涉：振幅为 2A
相消干涉：振幅为 0

8. Standing waves occur when both ends of a string are fixed. In that case, only waves which are motionless at the ends of the string can persist. There

are nodes, where the amplitude is always zero, and antinodes, where the amplitude varies from zero to the maximum value.
驻波由两列频率、振幅相同的相干波反向传播形成。
节点（Node）：振幅始终为零的点
波腹（Antinode）：振幅最大的点。
驻波频率由边界条件决定（如两端固定弦的基频和谐频）。

9. 衍射是波遇到障碍物或狭缝时，绕过边缘传播的现象。
衍射显著程度取决于障碍物尺寸与波长的比值：
障碍物或者孔的大小比波长小或者与波长差不多时，波的衍射现象明显。

【Example 1】
Under which of the following circumstances is the phenomenon of wave diffraction most prominent?
A. Water waves (with wavelength of 1 meter) passing through a sluice gate with a width of 10 meters.
B. Sound waves (with wavelength of 1 meter) passing through a circular hole with a diameter of 0.5 meters.
C. Light waves (with wavelength of 500 nanometers) passing through a slit with a diameter of 1 millimeter.
D. P-waves of earthquakes (with wavelength of 5 kilometers) passing through a fault with a width of 10 kilometers.

【辅助理解】
判断衍射是否显著，看障碍物尺寸和波长。选项 B 中声波波长 1 米，孔直径 0.5 米，接近波长，衍射最明显。选项 C 和 D 中波长远小于障碍物尺寸，衍射微弱；选项 A 中闸口宽度远大于波长，衍射不明显。故选 B

【Example 2】
A string with both ends fixed is 2 meters long and its wave velocity is 100 meters per second. What is its third harmonic frequency?
A. 25 Hz B. 50 Hz C. 75 Hz D. 100 Hz

【辅助理解】
驻波频率公式为

$$f_n = \frac{v}{\lambda_n} = n\frac{v}{2l} = nf_1 \quad , n = 1,2,3...$$

第三谐频对应 $n = 3$，代入得：

$$f_3 = 3 \times \frac{100}{2\times2} = 75\text{Hz}$$

选项 B 是第二谐频（n=2），选项 A 是基频（n=1），选项 D 为第四谐频（n=4）。故选 C

电荷性质与库仑定律

本节物理词汇：Electric charge (charge) 电荷
Coulomb's law 库仑定律
Quantized 量子化的
Electrostatic force 静电力
Nucleus 原子核

【基本思想】

1. 电荷分正负，同性相斥、异性相吸。

2. 电荷守恒，量子化（基本单位为电子电荷 $e = 1.602 \times 10^{-19}$）。

3. 描述点电荷间静电力：$F = k\dfrac{Q_1 Q_2}{r^2}$，方向沿连线。

4. 介电常数 ε_0 与 k 的关系：$k = \dfrac{1}{4\pi\varepsilon_0}$。

【Example 1】

Two point charges $Q_1 = +3\mu C$ and $Q_2 = -5\mu C$ are separated by 2 cm. Calculate the electrostatic force magnitude and direction.

【辅助理解】

Apply Coulomb's Law for magnitude; determine direction via charge interaction rules.

【Solution】

$$F = 9 \times 10^9 \times \frac{\left|3 \times 10^{-6} \times (-5) \times 10^{-6}\right|}{(0.02)^2} = 337.5N$$

Direction: Attractive (since charges are opposite), along the line connecting charges.

导体与绝缘体

本节物理词汇：Insulator/conductor 绝缘体/导体
Semiconductor 半导体
Induced charge 感应电荷，感生电荷
Electroscope 验电器

【基本思想】
1. 导体中电荷自由移动（如金属）；绝缘体中电荷几乎不移动。
2. 半导体介于两者之间。

【Example 1】
A neutral metal sphere is brought near a positively charged glass rod. Describe the induced charge distribution.

【辅助理解】
In conductors, electrons redistribute under external electric fields.

【Solution】
1. Electrons in the sphere are attracted to the rod's positive charge, accumulating on the near side (negative induced charge).
2. The far side becomes positively charged due to electron deficit.
3. Net charge remains zero (conservation of charge).

电场计算（点电荷），连续电荷分布的电场，导体内部电场，电场线应用

本节物理词汇：(Static) Electric field (静)电场
Electric field line 电场线
Permittivity 介电常数

【基本思想】

1. 定义：$\vec{E}=\frac{\vec{F}}{q}$，方向为正电荷受力方向。

2. 点电荷电场：$E=\frac{kQ}{r^2}$。
3. 连续电荷分布需积分计算电场。
4. 静电平衡时，导体内部电场为零；电荷分布在外表面，表面电场垂直。
5. 起于正电荷，止于负电荷；密度表场强，切线表方向。

【Example 1】

Calculate the electric field 3 cm from a $Q=+6\mu C$ point charge.

【辅助理解】
Use the point charge field formula; direction radiates outward for Q>0.

【Solution】

Formula: $E = k\frac{|Q|}{r^2}$

Substitution: $E = 9\times10^9 \times \frac{6\times10^{-6}}{(0.03)^2} = 6\times10^7 N/C$

Direction: Radially outward from Q.

【Example 2】
An infinite straight wire has linear charge density λ=2μC/m. Find the electric field 5 cm away.

【辅助理解】
Apply the infinite line charge formula derived from Gauss's Law.

【Solution】

Formula: $E = \frac{\lambda}{2\pi\varepsilon_0 r}$

Substitution: $E = \frac{2\times10^{-6}}{2\pi\times8.85\times10^{-12}\times0.05} \approx 7.19\times10^5 N/C$

Direction: Perpendicular to the wire (outward if λ>0).

【Example 3】
A hollow metallic shell carries net charge +Q. Determine the electric field inside and outside the shell.

【辅助理解】
Apply Gauss's Law and conductor properties.

【Solution】
Inside shell (r<R): E=0(electrostatic shielding)
Outside shell (r>R): $E = k\frac{Q}{r^2}$ (equivalent to point charge)

【Example 4】
Sketch field lines for two equal positive charges and determine the midpoint field.
【辅助理解】Field lines repel symmetrically; superposition principle applies.

【Solution】
Field lines emanate radially from each charge and repel at the midpoint.

At midpoint: $\vec{E}_{net} = 0$ (symmetrical cancellation)

电偶极子

基本物理词汇：Electric dipole 电偶极子
Electric dipole moment 电偶极矩
Polar 极性的

【基本思想】

1.等量异号电荷对，偶极矩 $\vec{p} = Q\vec{l}$ ；在均匀电场中受力矩 $\vec{\tau} = \vec{p} \times \vec{E}$ 。

【Example 1】

An electric dipole with a dipole moment $p = 4 \times 10^{-30} C \bullet m$ is placed in a uniform electric field $E = 10^5 N/C$. Find the maximum torque.

【辅助理解】

Torque is maximized when $\vec{p}$ is perpendicular to $\vec{E}$.

【Solution】

Maximum torque formula: $\tau_{\max} = pE$

$\tau_{\max} = 4 \times 10^{-25} N \bullet m$

Direction: Torque aligns $\vec{p}$ with $\vec{E}$ (right-hand rule).

电通量与高斯定理

本节物理词汇：

Gauss's law 高斯定律
Electric flux 电通量
Charge distribution 电荷分布
Surface integration 曲面积分
Electric potential 电势

【公式】

1. Electric flux（电通量）

与磁通量相似，是描述电场通过某一面积的电场线数量的物理量，是标量，单位是 $N\cdot m^2/C$，穿出高斯面的电通量为正，穿进高斯面的电通量为负。

$$\Phi_E = EA_\perp = EA_\perp = EA\cos\theta = \vec{E}\cdot\vec{A}$$

通过电场中某一面积的电场强度通量等于电场强度与该面积的矢量点积对该面积的积分。

概念拓展：可理解为电场线穿过某一个平面的“多少”，当 θ =0 时，即电场强度与面积垂直，电通量最大；当 θ =90°时，电场强度与面积平行，电通量为零，意味着没有电场线穿过该面积。

2. Gauss's law（高斯定律）

在真空中，通过任一封闭曲面的电场强度通量等于该曲面所包围的所有的电荷的代数和除以ε（与面外电荷无关，闭合的封闭曲面称为高斯面）。

高斯定理是库仑定律与电场强度叠加原理共同推断而来。

$$\Phi_e = \oint \vec{E}\cdot d\vec{A} = \frac{1}{\varepsilon_0}\sum_{i=1}^{n} q_i$$

对于单点电荷在半径为 r 的球面的球心：

$$\frac{Q}{\varepsilon_0} = \oint \vec{E}\cdot d\vec{A} = E(4\pi r^2)$$

对于单点电荷在不规则球面内（通用有效公式）：

$$d\Phi_e = \frac{q}{4\pi\varepsilon_0 r^2} dS\cos\theta = \frac{q}{4\pi\varepsilon_0}\frac{dS'}{r^2}$$

其中：$\frac{dS'}{r^2}$ 为立体角，$\frac{dS'}{r^2} = d\Omega$

$$\Phi_e = \frac{q}{4\pi\varepsilon_0}\oint d\Omega = \frac{q}{\varepsilon_0}$$

3. 高斯定律的应用

利用高斯定律求解往往需要该物体具有高度的对称性。
如均匀带电球面的电场强度：

(1) $0 < r < R$, $E = 0$

(2) $r > R$, $E = \frac{Q}{4\pi\varepsilon_0 r^2}$

如均匀带电球体的电场强度：

(1) $0 < r < R$, $\vec{E} = \frac{Q\vec{r}}{4\pi\varepsilon_0 R^3}$

(2) $r > R$, $E = \frac{Q}{4\pi\varepsilon_0 r^2}$

【Example 1】
Consider the two imaginary Gaussian surfaces, A_1,and A_2, shown in Fig. 22-10. The only charge present is the charge Q at the center of surface A_1. What is the net flux through each surface, A_1 and A_2?

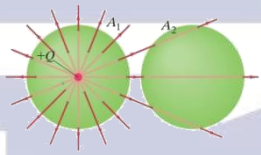


FIGURE 22–10 Example 22–2. Two gaussian surfaces.

【辅助理解】
此两面为高斯面，求电场强度通量。依据电场强度通量定义：穿过电场线的多少，穿入为负，穿出为正。A_1 平面射出的电场线，在 A_2 中全部穿进又穿出，故为 0

【Solution】
The surface A_1, encloses the charge +Q. By Gauss's law, the net flux through A_1, is then Q/ε_0. For surface A_2, the charge +Q is outside the surface. Surface A_2, encloses zero net charge, so the net electric flux through A_2 is zero, by Gauss' law. Note that all field lines that enter the volume enclosed by surface A_2 also leave it.

【Example 2】
Solid sphere（实心球体） of charge. An electric charge Q is distributed uniformly throughout a nonconducting sphere of radius r_0, Fig. 22-12. Determine the electric field (a) outside the sphere ($r > r_0$) and (b) inside the sphere ($r < r_0$).

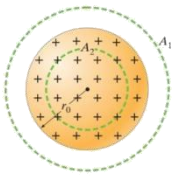


FIGURE 22–12 A solid sphere of uniform charge density. Example 22–4.

【辅助理解】
利用高斯定律的高对称性，该实心带点球体的电荷对称分布，所以所有点的电场也是对称的。故影响电场 E 大小的因素仅有 r。依据高斯定律进行求解。

【Solution】
(a) For our Gaussian surface we choose a sphere of radius r ($r > r_0$) labeled A_1, in Fig. 22-12. Since E depends only on r, Gauss's law gives, with $Q_{encl} = Q$.

$$\oint \vec{E} \cdot d\vec{A} = E(4\pi r^2) = \frac{Q}{\varepsilon_0}$$

or

$$E = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}$$

Again, the field outside a spherically symmetric distribution of charge is the same as that for a point charge of the same magnitude located at the center of the sphere.

(b) Inside the sphere, we choose for our Gaussian surface a concentric sphere of radius r ($r < r_0$), labeled A_2 in Fig. 22-12. From symmetry, the magnitude of E is the same at all points on A_2, and E is perpendicular to the Gaussian surface, so,

$$\oint \vec{E} \cdot d\vec{A} = E(4\pi r^2)$$

We must equate this to Q_{encl} / ε_0 where Q_{encl} is the charge enclosed by A_2. Q_{encl} is not the total charge Q but only a portion of it. We define the charge density, ρ_E, as the charge per unit volume ($\rho_E = dQ / dV$), and here we are given that ρ_E = constant. So the charge enclosed by the Gaussian surface A_2, a sphere of radius r, is

$$Q_{encl} = \left(\frac{\frac{4}{3}\pi r^3 \rho_E}{\frac{4}{3}\pi {r_0}^3 \rho_E}\right) Q = \frac{r^3}{{r_0}^3} Q$$

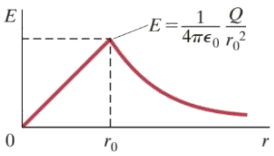


FIGURE 22–13 Magnitude of the electric field as a function of the distance *r* from the center of a uniformly charged solid sphere, Example 22–4.

Hence, from Gauss's law,

$$E(4\pi r^2) = \frac{Q_{encl}}{\varepsilon_0} = \frac{r^3}{{r_0}^3}\frac{Q}{\varepsilon_0}$$

or

$$E = \frac{1}{4\pi\varepsilon_0}\frac{Q}{{r_0}^3}r$$

Thus the field increases linearly with r, until $r = r_0$. It then decreases as $1 / r^2$? as plotted in Fig.22-13.

【Example 3】

Nonuniformly charged solid sphere（不均匀带电的球体）. Suppose the charge density of the solid sphere in Fig.22-12, Example 22-4, is given by $\rho_E = \alpha r^2$, where α is a constant. (a) Find α in terms of Q and r_0, the total charge on the sphere and its radius. (b) Find the electric field as a function of r inside the sphere.

【辅助理解】

将该球体进行微分，微分成厚度为 dr 的同心薄壳，（a）对这些球面进行积分得出电荷。（b）使用高斯定律的对称性，求出电场强度。

【Solution】

(a) A thin shell of radius r and thickness dr (Fig.22-14) has volume $dV = 4\pi r^2 dr$. The total charge is given by

$$Q = \int \rho_E dV = \int_0^{r_0} (\alpha r^2)(4\pi r^2) = 4\pi\alpha \int_0^{r_0} r^4 dr = \frac{4\pi\alpha}{5}{r_0}^5$$

Thus $\alpha = 5Q/4\pi{r_0}^5$.

(b) To find E inside the sphere at distance r from its center, we apply Gauss's law to an imaginary sphere of radius r which will enclose a charge

$$Q_{encl} = \int_0^r \rho_E dV = \int_0^r (\alpha r^2)4\pi r^2 dr = \int_0^r (\frac{5Q}{4\pi{r_0}^5}r^2)4\pi r^2 dr = Q\frac{r^5}{{r_0}^5}$$

By symmetry, E will be the same at all points on the surface of a sphere of radius r, so Gauss's law gives

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{encl}}{\varepsilon_0}$$

So

$$E(4\pi r^2) = Q\frac{r^5}{\varepsilon_0{r_0}^5}$$

Then

$$E = \frac{Qr^3}{4\pi\varepsilon_0{r_0}^5}$$

电势

本节物理词汇：

Electric potential 电势
Electrostatic potential energy 静电势能
Voltage 电压
Breakdown voltage 击穿电压
Cathode/anode 阴极/阳极
Equipotential surface 等势面
Electron volt 电子伏（特）

【公式】

1. 某点的电势是单个电荷在该点处的电势能：

$$V_{ba} = V_b - V_a = \frac{U_b - U_a}{q} = -\frac{W_{ba}}{q}$$

2. 两点间电势差：$V_{ba} = V_b - V_a = -\int_a^b \vec{E}d\vec{l}$

3. 真空中点电荷电势：（运用库仑定律）$V = \frac{1}{4\pi\epsilon_0}\frac{Q}{r}$

4. 等势：电势相同的线（等势线）或面（等势面）
5. 电偶极子的电势与 $1/r^2$ 成正比

【基本思想】

1. 势能的变化量等于保守力做功的负值。保守力做正功，势能减小；保守力做负功，势能增大。
2. 当电场分布不均匀时，需要利用积分求势能变化量，即

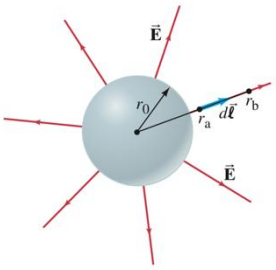
$$U_{ba} = U_b - U_a = -q\int_a^b \vec{E}d\vec{L}$$

根据电势与电势能的关系，可得

$$V_{ba} = V_b - V_a = -\int_a^b \vec{E}d\vec{l}$$

【Example 1】

Determine the potential at a distance r from the center of a charged conducting sphere of radius ro for (a) $r>r_0$ (b) $r=r_0$ (c) $r<r_0$. The total charge on the sphere is Q.



【辅助理解】

1. 对于带电导体球，电荷均匀分布在导体表面，导体内部电场强度为零。

2. 带电导体球外的电场大小公式 $E = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}$，利用此公式结合电势与电场的积分关系 $V_{ba} = V_b - V_a = -\int_a^b \vec{E}d\vec{l}$来计算电势。

3. 沿径向积分，并令无穷远处电势 $V = 0$ 。根据不同的 r 取值范围（$r>r_0$、$r = r_0$、$r < r_0$）分别进行积分计算。当 $r < r_0$ 时，要知道导体内部电场 $E = 0$ ，进而确定电势。

By 邱语

4. 球外电势等效于全部电荷集中在球心处点电荷产生的电势。

【Solution】

（a）当 $r > r_0$ 时

已知电势差公式 $V_{ba} = V_b - V_a = -\int_a^b \vec{E}d\vec{l}$，因为沿径向积分, r 与 $d\vec{l}$ 平行，$d\vec{l}$ 的方向与电场方向一致，所以 $\vec{E}d\vec{l} = E\,dr$ 。又已知导体球外电场 $E = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}$（$r > r_0$）。

令 $V_a = V$（所求电势），$r_a = r$，$V_b = 0$（无穷远处电势为 0 ），$r_b = \infty$ ，则 $V - 0 = -\int_r^\infty \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}dr$

对于 $\int_r^\infty \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}dr$，根据积分公式 $\int x^{-2}dx = -x^{-1} + C$，可得 $\int_r^\infty \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}dr = \frac{Q}{4\pi\epsilon_0}\int_r^\infty \frac{1}{r^2}dr = \frac{Q}{4\pi\epsilon_0}[-\frac{1}{r}]_r^\infty = \frac{Q}{4\pi\epsilon_0}[\frac{1}{r} - \frac{1}{\infty}]$

所以 $V = \frac{Q}{4\pi\epsilon_0 r}$

（b）当 $r = r_0$ 时

当 r 趋近于 r_0 时，在（a）中得到的球外电势公式 $V = \frac{Q}{4\pi\epsilon_0 r}$ 仍然适用（因为电势是连续的），将 $r = r_0$ 代入该公式，得 $V = \frac{Q}{4\pi\epsilon_0 r_0}$

（c）当 $r < r_0$ 时

因为导体内部电场强度 $E = 0$ ，根据 $V_b - V_a = -\int_a^b \vec{E}d\vec{l}$，对于导体内部任意两点，积分 $\int_a^b \vec{E}d\vec{l} = 0$，所以导体内部电势处处相等；而导体表面电势为 $V = \frac{Q}{4\pi\epsilon_0 r_0}$，所以导体内部电势与表面电势相同，

即 $V = \frac{Q}{4\pi\epsilon_0 r_0}$

【Example 2】

A flat ring of radius R_1 and outer radius R_2, carries a uniform surface charge density σ. Determine the electric potential at points along the axis (the x axis).

【辅助理解】

1. 题意：计算一个内半径为 R_1、外半径为 R_2、表面电荷密度均匀为 σ 的平面环在 x 轴上的电势
2. 将带电圆环分割成一个个宽度为 dr 的细圆环微元，把连续的电荷分布离散化处理。
3. 根据电势叠加原理，整个带电圆环在轴线上一点产生的电势等于各个微元在该点产生电势的总和。
4. 先求出单个细圆环微元在轴线上某点产生的电势 dV，再通过积分计算所有微元电势的累加，从而得到总电势 V 。

【Solution】

1. 将环形电荷分布分割为微小电荷元

dA = $2\pi r dr$

由于表面电荷密度为均匀的 σ，

该薄圆环的电荷量为 dq = σdA = $\sigma 2\pi r dr$

2. 计算电荷元到 x 轴上某点的距离

$$d = \sqrt{r^2 + x^2}$$

3. 利用点电荷的电势公式 V = $\frac{Q}{4\pi\epsilon_0 r}$

点电荷 dq 在距离 d 处产生的电势为 dV = $\frac{dq}{4\pi\epsilon_0 d}$

代入 dq 和 d 的表达式，得 dV = $\frac{\sigma 2\pi r dr}{4\pi\epsilon_0\sqrt{r^2+x^2}}$

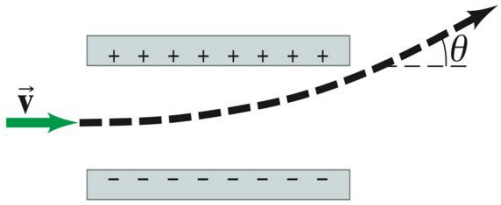
4. 积分求总电势 V： 对 r 从 R_1 到 R_2，积分，计算整个平面环在 x 轴上的电势

V = $\int_{R1}^{R2} \frac{\sigma 2\pi r}{4\pi\epsilon_0\sqrt{r^2+x^2}} dr = \frac{\sigma}{2\epsilon_0}\int_{R1}^{R2} \frac{r}{\sqrt{r^2+x^2}} dr$

换元, u = $r^2 + x^2$，则 du = 2rdr

V = $\frac{\sigma}{2\epsilon_0}[\sqrt{{R_2}^2 + x^2} - \sqrt{{R_1}^2 + x^2}]$

【Example 3】



An electron is accelerated horizontally from rest by a potential difference of 3500 V. It then passes between two horizontal plates 6.5 cm long and 1.3 cm apart that have a potential difference of 250V. At what angle θ will the electron be traveling after it passes between the plates? Assume the electric field between the plates is uniform.

【辅助理解】

题干翻译：一个电子从静止开始被电势差 V_1= 3500V 加速，然后进入两平行板之间的区域，两板电势差为 V_2= 250V，板间距为 d = 1.3 cm(= 0.013 m)。计算其加速后的初速度、在两板间的加速度、离开电场时的偏转角。

解题思路：

要求θ，可以运用正弦，即 $\tan\theta = \frac{v_y}{v_x}$， 那么目标是求出 v_x 和 v_y

① 为求 v_x，注意到题干第一句话“电子被 3500V 的电势差水平加速”，由此根据

U = $qV_1 = \Delta K = \frac{1}{2} mv_x^2$ 可求得 v_x

② 为求 v_y，由于 $v_y = \frac{qE}{m}t$（加速度 a 是在极板间电场里作用下产生，根据牛顿第

二定律 F = ma，得 a = $\frac{qE}{m} = \frac{qV_2}{md}$ ）

又因为 x = $v_0 t$，所以 t = $\frac{L}{v_x}$

注意，本题没有条件说明电子刚好从极板末端离开。

【Solution】

1. 求电子加速后的初始水平速度

根据动能定理，电子在电势差 V_1= 3500V 下加速时，电场力做功等于动能变化。因为电子初始静止，所以

$$qV_1 = \Delta K = \frac{1}{2} mv_x^2$$

其中 $q = 1.6 \times 10^{-19}$ C（电子电荷量），$m = 9.11 \times 10^{-31}$ kg（电子质量）。$V_x = \sqrt{\frac{2qV1}{m}}$

$$\approx 3.51\times10^{7} \text{ m/s}$$

2. 求电子在两板间的加速度

两板电势差 $V_2 = 250$ V，板间距 $d = 1.3$ cm = 0.013 m，

电场强度 $E = \frac{V_2}{d}$，电子所受电场力 $F = qE$，

牛顿第二定律得加速度：$a = \frac{qE}{m} = \frac{qV_2}{md} \approx 3.37 \times 10^{15}$ m/s^2

3. 求电子出射时的速度偏转角 θ

(1) 求电子在板间的运动时间

如辅助理解中所述，$t = \frac{L}{v_x}$

(2) 求电子离开板时的竖直速度

竖直方向初速度为零，末速度 $v_y = at$

(3) 速度偏转角的正切值为：

$\tan\theta = \frac{v_y}{v_x} = \frac{aL}{v_x^2} \approx 0.117$

$\theta = \tan^{-1} 0.117 \approx 0.175$ rad

电流和电阻

本届物理词汇：

electric current 电流
Ohm's Law 欧姆定律
Resistance 电阻
Resistivity 电阻率
Resistor 电阻器
Direct Current (DC) 直流电
Alternating Current (AC) 交流电
Current density 电流密度
Drift velocity 漂移速度
Electric power 电功率

【概念】

电流：单位时间内通过导体横截面的电荷量，方向定义为正电荷移动方向。
电阻率：材料本身的电学特性，与形状无关，单位Ω·m。
交流电：大小和方向周期性变化的电流（如家用电源）。
电流密度：单位面积上的电流强度，反映电流分布均匀性。
漂移速度：电荷在电场作用下定向移动的平均速度。
电功率：单位时间电能转化为其他形式能量的速率。

【公式】

1. 电流定义：$I = \frac{dQ}{dt}$ （电荷量是标量，电流方向由电压方向决定）
2. 欧姆定律：$V = IR$
3. 电阻与电阻率关系：$R = \rho \frac{l}{A}$
4. 电功率公式：$P = IV = I^2 R = \frac{V^2}{R}$

   当电流通过电阻时，电能转化为热能
5. 交流电表达式：

   $V = V_0 \sin(\omega t) = V_0 \sin(2\pi f t)$ $\quad I = I_0 \sin(2\pi f t)$
6. 电流面密度：

   $j = \frac{I}{A} = nev_d$（n 为单位体积自由电荷数，e 为电子电荷量，v_d 为漂移速度）

【基本思想】

1. 欧姆定律

金属导体在温度不变时遵循欧姆定律，V-I 图像为过原点的直线。
半导体、气体放电等非线性元件不满足欧姆定律。

2. 电阻率：

电阻率是材料的固有属性，与形状无关。
温度升高时，金属电阻率增大（原子振动加剧阻碍电子运动），半导体电阻率减小（更多电子被激发参与导电）。

3. 电流的微观解释

自由电子在导体中做无规则热运动，叠加电场引起的定向漂移
漂移速度虽小，但因电子密度极高，宏观电流仍显著。
4. 交流电的有效值
交流电的有效值定义为与直流电产生相同热效应的电流值
【Example 1】
A copper wire has length l = 2.0 m and diameter d = 1.0 mm. Given copper's resistivity $\rho = 1.68 \times 10^{-8}\ \Omega\cdot m$, calculate its resistance.
【辅助理解】
目标：用电阻公式 $R = \rho\frac{l}{A}$ 计算导线电阻
计算横截面积: 导线是圆柱形，横截面积 $A = \pi r^2$
注意单位转换
代入电阻公式求解
【Solution】
$r = 0.5 \times 10^{-3}$ m
$A = \pi r^2 = \pi\, 0.5^2 = 7.85 \times 10^{-7}\ m^2$
$R = \rho\frac{l}{A} = 1.68 \times 10^{-8} \times \frac{2.0}{7.85 \times 10^{-7}} \approx 0.043\ \Omega$

【Example 2】
A uniformly stretched wire with original length l_1 and resistance R_1. If stretched to $l_2 = 2\, l_1$ while maintaining volume, find the new resistance R_2.
【辅助理解】
分析导线拉伸后电阻变化，理解体积不变条件的影响。
体积不变：拉伸后导线变长，但总材料体积 V = A l 不变。
电阻公式：电阻与长度成正比，与横截面积成反比
(想象将橡皮筋拉长 2 倍，其横截面积会缩小为原来的一半)
长度加倍使电阻加倍，面积减半使电阻再加倍，总电阻变为 4 倍。
【Solution】
$V = A\,l$, $R = \rho\frac{l}{A}$, $l_2 = 2\, l_1$ Thus $R_2 = 4R_1$
【Example 3】
A copper wire with current density $j = 6.2\times10^5\ A/m^2$ and cross-sectional area $A = 2.0\times10^{-6}\ m^2$. Calculate the current and electron drift velocity (given $n = 8.5 \times10^{28}\ m^{-3}$, $e = 1.6 \times10^{-19}$C) .
【辅助理解】
电流密度 j 是单位面积的电流，因此总电流 I = j A
漂移速度计算：公式 $j = \frac{I}{A} = n\, e\, v_d$ 变形得 $v_d = \frac{j}{ne}$
物理意义：电子越多或电荷量越大，相同电流密度下漂移速度越小。
【Solution】
$I = 6.2\times10^5\ A/m^2 \times 2.0\times10^{-6}\ m^2 = 1.24$ A
$v_d = \frac{j}{ne} \approx 4.5 \times10^{-5}$ m/s

基尔霍夫定律

本节物理词汇:

EMF (Electromotive Force) 电动势
Terminal Voltage 终端电压
Resistors in Series 串联电阻
Resistors in Parallel 并联电阻
Kirchhoff's Rules 基尔霍夫定律
RC Circuits RC 电路
Electric Hazards 电击危险

【概念】

1. 电动势ε：电池或发电机提供的能量转化能力，单位为伏特（V）。实际电池存在内阻 r ，导致终端电压小于电动势。
2. 终端电压：电池实际输出的电压，计算公式 $V = \varepsilon - Ir$ ，其中 I 为电流，r 为内阻。
3. Kirchhoff's Rules 基尔霍夫定律：
（1）节点定律：流入节点的电流总和等于流出电流总和。
（2）环路定律：闭合回路中电势变化的代数和为零。
4. RC Circuits RC 电路：含电阻和电容的电路，充放电过程遵循指数规律，时间常数 $\tau = RC$。
5. Electric Hazards 电击危险：人体通过电流超过 10mA 可能导致肌肉失控，100mA 以上可能致命。

【公式】

1. 终端电压：$V_{terminal} = \varepsilon - Ir$
   ε 为电动势，r 为内阻
2. 串联等效电阻：$R_{eq} = R_1 + R_2 + R_3 + \ldots$
3. 并联等效电阻：$\frac{1}{R_{ep}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \ldots$
4. RC 电路充放电时间常数：$\tau = RC$

【基本思想】

1. 电动势与内阻：
理想电池 $V_{terminal} = \varepsilon$，实际电池因内阻分压导致输出降低。
(类比**：水泵提供的水压（电动势）因管道摩擦（内阻）而降低)
2. 基尔霍夫定律应用：
(1) 节点定律：电流分配遵循“进=出”
(2) 环路定律：沿闭合路径电压升↑ = 电压降↓（类似能量守恒）。
3. RC 电路特性：
充电时电容电压逐渐上升，电流逐渐下降；放电时相反。
时间常数 τ 表示充放电速度， τ 越大过程越慢。

【Example 1】

A battery with EMF $\varepsilon = 12.0$ V and internal resistance $r = 0.5\ \Omega$ is connected to an external resistor $R = 65.0\ \Omega$. Find:
(a) Circuit current;

(b) Terminal voltage;
(c) Power dissipated in R and r .
【辅助理解】
题目翻译：一电池电动势 $\varepsilon = 12.0$ V，内阻 $r = 0.5\ \Omega$，外接 $R = 65.0\ \Omega$的电阻。求：(a) 电路电流；(b) 终端电压；(c) 外电阻和内阻的功率。

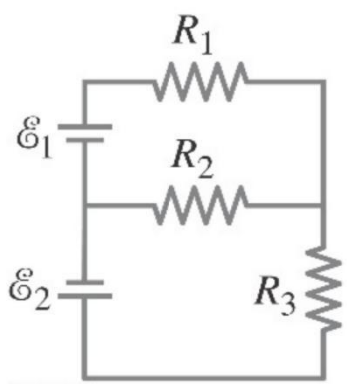
1. 总电阻 = 外阻 + 内阻 $\rightarrow I = \frac{\varepsilon}{R + r}$

2. 终端电压 = 电动势-内阻压降 $\rightarrow V_{terminal} = \varepsilon - Ir$
3. 功率公式 $P = I^2 R$
【Solution】

(a) $R_{total} = 65.0\ \Omega + 0.5\ \Omega = 65.5\ \Omega$， $I = \frac{E}{R_{total}} \approx 0.183$ A

(b) $V_{terminal} = \varepsilon - Ir = 12.0\text{ V} - (0.183\text{ A}\times 0.5\ \Omega) \approx 11.91$V
(c)外电阻功率 $P_R = (0.183)^2 \times 65.0\text{ W} \approx 2.18$ W
内阻功率 $P_r = (0.183)^2 \times 0.5\text{ W} \approx 0.017$ W

【Example 2】
Determine the magnitudes and directions of the currents in each resistor shown in the figure. The batteries have emfs of $\varepsilon_1 = 9.0$ V and $\varepsilon_2 = 12.0$ V and the resistors have values of $R_1 = 25\ \Omega$, $R_2 = 48\ \Omega$, and $R_3 = 35\ \Omega$
(a) Ignore internal resistance of the batteries.
(b) Assume each battery has internal resistance $r = 1.0\ \Omega$
【辅助理解】
1. 关键概念：基尔霍夫定律
节点处流入电流之和等于流出电流之和，任一闭合回路的电压降之和等于电动势之和。



（1）电路假设：
假设电路为双环路结构，其中 ε_1与 R_1、R_2组成环路 1，ε_2 与 R_2 、 R_3组成环路 2，共享电阻 R_2。
（2）电流方向定义：
I_1 ：从 ε_1正极经 R_1 到节点 A，再经 R_2到节点 B，返回 ε_1负极。
I_2 ：为通过 R_2 的总电流，方向由叠加结果决定。
I_3 ：从ε_2正极经 R_3到节点 B，再经 R_2 到节点 A，返回 ε_2 负极。
（3）. 基尔霍夫定律应用：
①节点定律：在节点 A，$I_1 = I_2 + I_3$
②环路定律：
环路 1： $\varepsilon_1 - I_1 R_1 - I_2 R_2 = 0$
环路 2： $\varepsilon_2 - I_3 R_3 - I_2 R_2 = 0$
2. 欧姆定律：V=IR，用于计算电阻两端电压。
3. 内阻影响：(b)问需将电池内阻视为串联在电路中的电阻
【Solution】
（a）忽略电池内阻

$$\begin{cases} 9-25I_1-48I_2=0 \\ 12-35I_3-48I_2=0 \\ I_1=I_2+I_3 \end{cases}$$

解得

$I_1 \approx 0.339$ A 方向：从ε_1正极经 R_1 到节点 A

$I_2 \approx 0.011$ A 方向：从节点 A 经 R_2 到节点 B

$I_3 \approx 0.328$ A 方向：从ε_2正极经 R_3 到节点 B

（b）考虑电池内阻 r =1.0 Ω

修正电路模型

每个电池串联内阻 r =1 Ω，即：

ε_1 支路(不包括 R_2)总电阻：$R_1 + r = 25 + 1 = 26\Omega$

ε_2 支路(不包括 R_2)总电阻：$R_3 + r = 35 + 1 = 36\ \Omega$

则 $$\begin{cases} 9-26I_1-48I_2=0 \\ 12-36I_3-48I_2=0 \\ I_1=I_2+I_3 \end{cases}$$

解得 $I_1 \approx 0.331$ A 方向：从ε_1正极经 R_1 到节点 A

$I_2 \approx 0.0085$ A 方向：从节点 A 经 R_2 到节点 B

$I_3 \approx 0.322$ A 方向：从ε_2正极经 R_3 到节点 B

磁学

本节物理词汇：

Magnet 磁体
Magnetic field lines 磁场线
North and South poles 磁北南极
Magnetic dipole moment 磁矩
Hall effect 霍尔效应
Mass spectrometer 质谱仪
Tesla 特斯拉 磁场单位
Lorentz force 洛伦兹力
Right-hand rule 右手定则
Current loop 电流环
Torque on a current loop 电流环扭矩
Uniform magnetic field 均匀磁场
Drift velocity 漂移速度
Potential difference Hall emf 电势差 霍尔电动势
Charge carriers 电荷载流子
Circular motion of charged particles 带电粒子圆周运动
Velocity selector 速度选择器
Crossed electric and magnetic fields 交叉电磁场

【公式】

1.安培力公式（Force on a current-carrying wire）

$$F = I\ell B\sin\theta$$

电流为 I、长度为ℓ的导线在磁场 B 中受到的力，方向由右手定则判定

2.安培力矢量形式（Vector form of Ampere's force law）

$$\vec{F} = I\vec{\ell} \times \vec{B}$$

3.洛伦兹力公式（Lorentz force law）

$$F = q\vec{v} \times \vec{B}$$

电荷 q 以速度 v 在磁场 B 中受到的力

4. 电流环扭矩公式（Torque on a current loop）

$$\tau = NIAB\sin\theta$$

N 匝面积为 A 的载流线圈在磁场 B 中受到的扭矩

5.磁矩定义（Definition of magnetic moment）

$$\vec{\mu} = NI\vec{A}$$

电流环的磁矩，方向由右手定则判定

6.磁矩势能公式（Potential energy of magnetic moment）

$$U = -\mu B\cos\theta = -\vec{\mu}\cdot\vec{B}$$

7.霍尔效应平衡条件（Hall effect equilibrium condition）

$$eE_H = ev_dB$$

8.霍尔电动势公式（Hall EMF formula）

$$\xi_H = E_Hd = v_dBd$$

9.电流与漂移速度关系（Current and drift velocity relation）

$$I = neAv_d$$

10.速度选择器条件（Velocity selector condition）

$$\vec{F}_E = \vec{F}_B$$

$$E = vB$$

电场与磁场平衡时，粒子速度 v 满足的条件

11.功率与动能关系（Power and kinetic energy relation）

$$P = \frac{dW}{dt} = \vec{F}\cdot\vec{v} = q(\vec{v}\times\vec{B})\cdot\vec{v} = 0$$

洛伦兹力不做功的数学表达（因力始终垂直于速度）

【Example 1】

A proton moves at a speed of $v = 4.0\times10^6\ m/s$ in a direction 30° above the horizontal in a magnetic field.

The magnetic field exerts a force of $F = 6.0\times10^{-14}\ N$ toward the east on the proton.

When the proton moves horizontally along a certain axis, the magnetic force is zero. （$q = 1.6\times10^{19}\ C$）

【辅助理解】

粒子水平运动时无力，说明磁场方向在水平方向；用洛伦兹力公式和右手定则来求解磁场方向和大小。

【Solution】

由于水平运动时不受力，说明磁场方向与水平运动方向平行（即沿南北方向）。

则实际有效速度为垂直分量 v_y，因此：

$$B = \frac{F}{qv_y} = \frac{F}{qv\sin 30} = \frac{3}{32}T$$

Direction of B: Vertically upward

【Example 2】
A proton with mass $m = 1.0 \times 10^{-27}\,\mathrm{kg}$ and charge $q = 1.6 \times 10^{19}\,\mathrm{C}$ enters a uniform magnetic field perpendicularly at speed v=2.0×10^5m/s.
It follows a semicircular path of radius $r = 10\,\mathrm{cm}$ and hits a detector placed directly opposite its entry point.
(1) What must the magnetic field strength B be?
(2) If the magnetic field is increased to twice that value, will the proton still hit the detector? Why or why not?

【辅助理解】
磁场可以使带电粒子做圆周运动。同一粒子粒子运动半径由速度和磁场决定。

【Solution】
(1)

$$qvB = \frac{mv^2}{R}$$

$$B = \frac{mv}{qR} = 0.0125\mathrm{T}$$

(2)

$$R = \frac{mv}{qB}$$

$$R \propto \frac{1}{B}$$

$$R' = 0.5R = 5\mathrm{cm}$$

The proton will bend more sharply and not reach the detector

【Example 3】
In a velocity selector, electric field $E = 3.0 \times 10^3\,\mathrm{V/m}$ points upward, and magnetic field $B = 0.020\mathrm{T}$ points into the page.
(1) Find the speed v of a particle that passes through without deflection.
(2) If a negative charge enters with speed v, what is the direction of its net force?
【辅助理解】
本题为速度选择器原理，$v = \frac{E}{B}$ 。

【Solution】
(1) No deflection means:

$$qE = qvB \Rightarrow v = \frac{E}{B} = 1.5\ \times 10^5\mathrm{m/s}$$

(2) For a negative charge, electric force is downward, magnetic force (right-hand rule, reversed for negative) is also downward → net force points downward → deflected down.

【Example 4】

Given that an electron in a hydrogen atom orbits the proton in a circular path of radius r, express the magnitude of its orbital magnetic dipole moment using symbols.

【辅助理解】
电子绕质子运动相当于一个带电环形电流，产生磁偶极矩。

【Solution】
Find the orbital period and equivalent current
The period of revolution is:

$$T = \frac{2\pi r}{v}$$

The equivalent current due to the electron's motion is:

$$I = \frac{e}{T} = \frac{ev}{2\pi r}$$

The area of the circular orbit is:

$$A = \pi r^2$$

The magnetic dipole moment is:

$$\mu = I \cdot A = \frac{evr}{2}$$

磁场

本节物理词汇：

Solenoid 螺线管
Toroid 环形线圈
Ferromagnetism 铁磁性
Hysteresis 磁滞现象
Paramagnetism 顺磁性
Diamagnetism 抗磁性
Permeability 磁导率
Magnetic susceptibility 磁化率
Coaxial cable 同轴电缆
Magnetic field 磁场
Coulomb 库仑
Domain 磁畴
Electromagnet 电磁铁

【基本思想】

1.Ampere's Law 安培定律

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{encl}}$$

2.Magnetic Field Due to a Straight Wire 无限长直导线磁场

磁场大小与距离成反比：

$$B = \frac{\mu_0 I}{2\pi r} \quad (r \geq R)$$ （导线外）

导线内部（半径 r < R）磁场随距离线性增加：

$$B = \frac{\mu_0 I r}{2\pi R^2}$$

3.Force between Two Parallel Wires 平行导线间作用力

$$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d}$$

方向：同向电流吸引，反向电流排斥。

4.Magnetic Field of a Solenoid and a Toroid 螺线管与环形线圈的磁场

螺线管内部磁场（近似均匀），n 为单位磁场匝数

$$B = \mu_0 n I$$

环形线圈内部磁场（沿环形路径分布），N 为总匝数：

$$B = \frac{\mu_0 N I}{2\pi r}$$

5. Biot-Savart Law 毕奥-萨伐尔定律

$$\vec{B} = \frac{\mu_0 I}{4\pi} \int \frac{d\vec{l} \times \hat{r}}{r^2}$$

6. 磁性材料的分类：
铁磁性（如铁、镍）：存在磁畴，强磁化且保留剩磁。
顺磁性（如铝、氧气）：弱磁化，与磁场同向。
抗磁性（如铜、水）：弱磁化，与磁场反向。

磁滞现象：铁磁材料的磁化强度依赖外部磁场历史，形成磁滞回线。剩磁 B_r和矫顽力 H_c是关键参数。

【Example 1】
Which of the following statements is correct regarding the application of Ampere's Law?

A. The magnetic field outside an infinitely long straight wire is $B = \frac{\mu_0 I}{2\pi r}$, but the magnetic field inside the wire is zero.
B. The magnetic field outside the outer conductor of a coaxial cable is solely determined by the current in the outer conductor.
C. The magnitude of the magnetic field inside a solenoid is $B = \mu_0 n I$, where n is the total number of turns.
D. The magnetic field of a circular coil (circular solenoid) is uniformly distributed in the external region.

【辅助理解】

A 错误，导线内部的磁场为 $B = \frac{\mu_0 I r}{2\pi R^2}$ （R 为导线半径）。

B 正确，根据安培定律，同轴电缆外导体外的总电流为内外导体电流的代数（反向），若总电流为零则磁场为零。
C 错误，n 是单位长度的匝数，而非总匝数。
D 错误，环形线圈的磁场完全集中于环内，外部磁场为零。
【solution】B

【Example 2】
Which of the following descriptions about magnetic materials is incorrect?
A. The larger the area of the hysteresis loop of ferromagnetic materials, the greater the residual magnetism and coercive force usually are.
B. The magnetic susceptibility of paramagnetic materials X_m is negative.
C. Diamagnetic materials will slightly weaken the total magnetic field in the

external magnetic field.
D. The magnetic permeability μ of ferromagnetic materials is much greater than the permeability of vacuum μ_0 .

【辅助理解】
顺磁性材料的磁化率 X_m为正值（$\mu > \mu_0$），而抗磁性材料的 X_m为负值。

【solution】B

【Example 3】
An infinitely long coaxial cable is composed of an inner conductor (radius R_1) and an outer conductor (inner radiusR_2, outer radius R_3). The inner conductor carries a uniformly distributed current I_0 (direction perpendicular to the paper outward), and the outer conductor carries an opposite uniformly distributed current I_0.
(a) Using Ampere's law, derive the magnetic induction intensity B for the following regions:
（1）$R<R_1$
（2）$R_2<R<R_3$
(b) If the current of the outer conductor is changed to be in the same direction as that of the inner conductor and still has the same magnitude I_0, analyze the change in the magnetic field at point $R>R_3$.

【辅助理解】
（a）针对内导体内部（$R<R_1$），利用电流均匀分布的特性，选取对称圆形路径，计算包围电流并代入安培定律，得到磁场随半径线性增大的表达式
对于外导体内部（$R_2<R<R_3$），需叠加内导体的全部正向电流与外导体的部分反向电流，通过比例截面积计算实际包围电流，最终导出磁场随半径减小的关系
（b）当外导体电流方向与内导体同向时，外部总电流加倍，磁场增强为原值的两倍

【solution】
(a)
(1) As $\oint B \cdot dl = \mu_0 I_{encl}$

$B \cdot 2\pi R = \mu_0(\frac{I_0}{\pi R_1^2} \cdot \pi R^2)$,then $B = \frac{\mu_0 I_0 R}{2\pi R_1^2}$

(2) $I_{encl} = I_0 - \frac{\pi(R^2-R_2^2)}{\pi(R_3^2-R_2^2)} I_0 = I_0(1 - \frac{R^2-R_2^2}{R_3^2-R_2^2})$

$B \cdot 2\pi R = \mu_0 I_{encl}$,then $B = \frac{\mu_0 I_0}{2\pi R}(\frac{R_3^2-R^2}{R_3^2-R_2^2})$

(b)

$I_0 + I_0 = 2I_0$

$B \cdot 2\pi R = \mu_0 \cdot 2I$,then $B = \frac{\mu_0 I}{\pi R}$

电磁感应和法拉第定律

本节物理词汇：

Induced EMF 感生电动势
Faraday's Law of Induction 法拉第电磁感应定律
Lenz's Law 楞次定律
Electric Generator 发电机
Back EMF 反电动势
Eddy current 涡电流
Transformer 变压器
Magnetic Flux 磁通量
Inductance 电感

【公式】

1. Magnetic Flux 磁通量

$$\Phi_B = \int \vec{B} \cdot d\vec{A}$$

磁通量与 Ch22 高斯定律中的电场强度通量的概念相似，为这个平面内穿过的磁场线的“多少”。

磁通量与穿过回路的磁场线总数成正比，单位为韦伯（Wb）。

$$1\text{Wb} = 1\text{T} \cdot \text{m}^2$$

由上述积分公式可得出：

$$\Phi_B = BA\cos\theta$$

2. Faraday’s Law 法拉第电磁感应定律公式

$$\xi = -N\frac{d\Phi_B}{dt}$$

感应电动势与磁通量的变化率成正比。

注意：N 为线圈匝数。计算时不要忽略线圈匝数！！！

变化的磁通量产生电场：

$$\oint \vec{E} \cdot d\vec{l} = -\frac{\Phi_B}{dt}$$

磁通量的变化率越大，产生的电场越强。

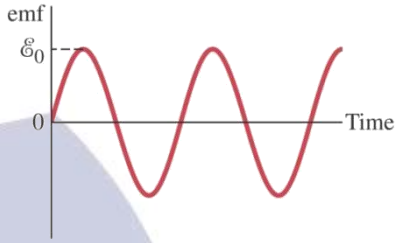
3. EMF Induced in a Moving Conductor 动生电动势公式

$$\xi = \frac{d\Phi_B}{dt} = \frac{B\,dA}{dt} = \frac{B\,l\,v\,dt}{dt} = B\,l\,v$$

B 为磁感应强度，l 为导体棒长度，v 为导体棒运动速度。
注意：适用的前提条件是：B, l, v 相互垂直，如果不垂直，需要乘$\cos\theta$（即取垂直分量）

4. Electric Generators 发电机感应电动势

$$\xi = NBA\omega \sin \omega t = \xi_0 \sin \omega t$$



其中：$\xi_0 = NBA\omega$ 为感应电动势的峰值

ω为线圈转动的角速度，A 为线圈面积。

5. Transformers 变压器

两边电压与线圈匝数成正比，两边电流与线圈匝数成反比，两边功率相等（理想情况下不计损耗），即：

$$\frac{V_S}{V_P} = \frac{N_S}{N_P}$$

V_s, V_P 分别为副线圈，原线圈的电压，N_S, N_P 分别为副线圈、原线圈的匝数

$$\frac{I_S}{I_P} = \frac{N_P}{N_S}$$

变压器的运用：远距离输电

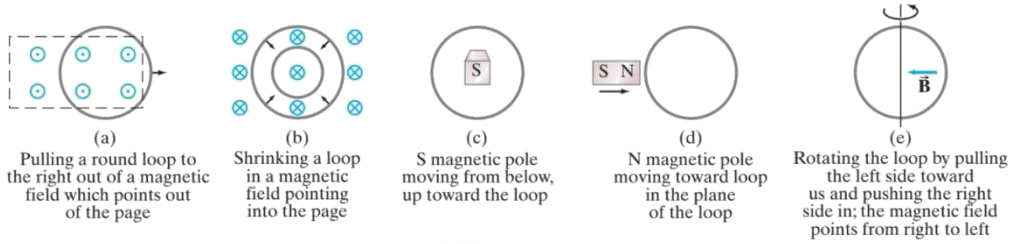
远距离输电降低能耗的方法：
（1）减小电线电阻（如选用电阻率小的电线）
（2）减小输电电流（如采用特高压输电）

6. Induced EMF(Induced Electromotive Force) 感生电动势

如何判断感生电动势方向？楞次定律（Lenz’s Law）
Lenz's Law states that the direction of the induced electromotive force (emf) is always such that the current it causes to flow produces a magnetic field that opposes the original change in magnetic flux that induced it.

感应电动势总是阻碍引起它的原磁通量的变化。

举例子：如用磁铁从下往上穿线圈，此时磁通量变大，若要阻碍这种趋势，需要让产生的电流产生的磁场减少磁通量（阻碍磁通量增大），即可判断电流的方向。



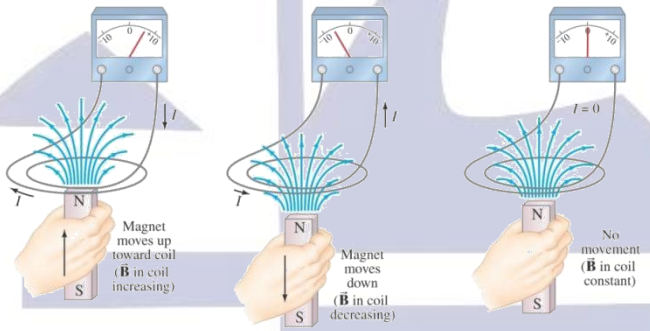
(a) Pulling a round loop to the right out of a magnetic field which points out of the page

(b) Shrinking a loop in a magnetic field pointing into the page

(c) S magnetic pole moving from below, up toward the loop

(d) N magnetic pole moving toward loop in the plane of the loop

(e) Rotating the loop by pulling the left side toward us and pushing the right side in; the magnetic field points from right to left



【Example 1】

A loop of wire in a magnetic field. A square loop of wire of side l = 5.0cm is in a uniform magnetic field B=0.16T. What is the magnetic flux in the loop

(a) when B is perpendicular to the face of the loop and

(b) when B is at an angle of 30° to the area A of the loop?

(c) What is the magnitude of the average current in the loop, if it has a resistance of 0.012 Ω, while it is rotated from position (b) to position (a) in 0.14s?

【辅助理解】

计算磁通量多少，直接使用磁通量公式，但需要注意角度问题，还有线圈匝数问题（本题不涉及线圈匝数问题，但后续题目需要注意）

【Solution】

The area of the coil is $A = l^2 = (5.0 \times 10^{-2})^2 = 2.5 \times 10^{-3}\mathrm{m}^2$ and the direction of $\vec{A}$ is perpendicular to the face of the loop (Fig. 29-3).

(a)$\vec{B}$ is perpendicular to the coil's face, and thus parallel to $\vec{A}$ (Fig. 29-3), so

$$\Phi_B = \vec{B} \cdot \vec{A} = BA\cos 0^\circ = (0.16\mathrm{T})(2.5 \times 10^{-3}) = 4.0 \times 10^{-4}\mathrm{T} \cdot \mathrm{m}^2$$

(b)The angle θ between $\vec{B}$ and $\vec{A}$ is 30°, and cos 30° = 0.866, so

$$\Phi_B = \vec{B}\cdot\vec{A} = BA\cos\theta = (0.16T)(2.5\times10^{-3})\cos 30^\circ = 3.5\times10^{-4}T\cdot m^2$$

(c)The magnitude of the induced emf (Eq. 29-2a) during the 0.14-s time interval is

$$\xi = \frac{\Delta\Phi_B}{\Delta t} = \frac{(4.0\times10^{-4}T\cdot m^2) - (3.5\times10^{-4}T\cdot m^2)}{0.14s} = 3.6\times10^{-4}V$$

Before and after the loop rotates, when it is at rest, the emf is zero. The current in the wire loop (Ohm's law) while it is rotating averages

$$I = \frac{\xi}{R} = \frac{3.6\times10^{-4}V}{0.012\Omega} = 0.030A = 30mA$$

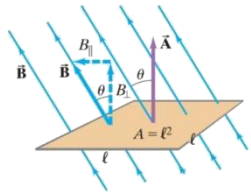


FIGURE 29–3 Determining the flux through a flat loop of wire. This loop is square, of side ℓ and area $A = \ell^2$.

【Example 2】

Pulling a coil from a magnetic field. A 100-loop square coil of wire, with side l = 5.00cm and total resistance R= 100 Ω, is positioned perpendicular to a uniform magnetic field B = 0.600T, as shown in Fig.29-11It is quickly pulled from the field at constant speed (moving perpendicular to B) to a region where B drops abruptly to zero. At t = 0, the right edge of the coil is at the edge of the field. It takes 0.100s for the whole coil to reach the field-free region. Determine (a) the rate of change in flux through one loop of the coil, and (b) the total emf and current induced in the 100-loop coil. (c) How much energy is dissipated in the coil? (d) What was the average force required (F_{ext})?

【辅助理解】

依据楞次定律找方向，然后依据磁通量的变化量找产生的电流大小，然后依据安培定律去求出受力大小。

【Solution】

(a) The coil's area is $A = l^2 = (5.00\times10^{-2}m)^2 = 2.50\times10^{-3}m^2$. The flux through one loop is initially $\Phi_B = BA = (0.600T)(2.50\times10^{-3}m^2) = 1.50\times10^{-3}T\cdot m^2$. After 0.100s, the flux is zero. The rate of change in flux is constant (because the coil is square), and for one loop is equal to

$$\frac{\Phi_B}{\Delta t} = \frac{0 - (1.50\times10^{-3}T\cdot m^2)}{0.100s} = -1.50\times10^{-2}T\cdot m^2/s$$

(b) The emf induced (Eq.29-2) in the 100-loop coil during this 0.100-s interval is

$$\xi = -N\frac{\Delta\Phi_B}{\Delta t} = -(100)(-1.50\times10^{-2}T\cdot m^2/s) = 1.50V$$

The current is found by applying Ohm's law to the 100-Ω coil:

$$I = \frac{\xi}{R} = \frac{1.50V}{100\Omega} = 1.50 \times 10^{-2}A = 15.0mA$$

By Lenz's law, the current must be clockwise to produce more $\vec{B}$ into the page and thus oppose the decreasing flux into the page.

(c)The total energy dissipated in the coil is the product of the power $(= IR^2)$ and the time:

$$E = Pt = I^2Rt = (1.50 \times 10^{-2}A)^2(100\Omega)(0.100s) = 2.25 \times 10^{-3}J$$

(d)We can use the result of part (c) and apply the work-energy principle: the energy dissipated E is equal to the work W needed to pull the coil out of the field (Chapters 7 and 8). Because $W = \overline{F_{ext}}\, d$ where d= 5.00 cm, then

$$\overline{F_{ext}} = \frac{W}{d} = \frac{2.25 \times 10^{-3}J}{5.00 \times 10^{-2}m} = 0.0450N$$

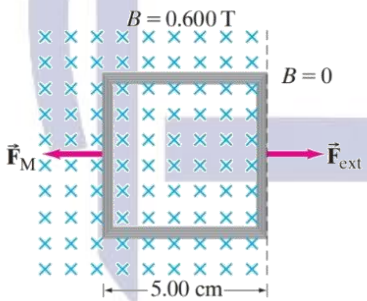


FIGURE 29–11 Example 29–5. The square coil in a magnetic field $B = 0.600\,T$ is pulled abruptly to the right to a region where $B = 0$. (The forces shown are discussed in the alternate solution at the end of Example 29–5.)